

Score Forecasts

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Load in truth data and forecasts

We load in the hospitalization truth data and our forecasts. (Note: The load forecast parentheses function wasn't working, so alternate code is written to achieve the same output.)

Versioned Truth Results

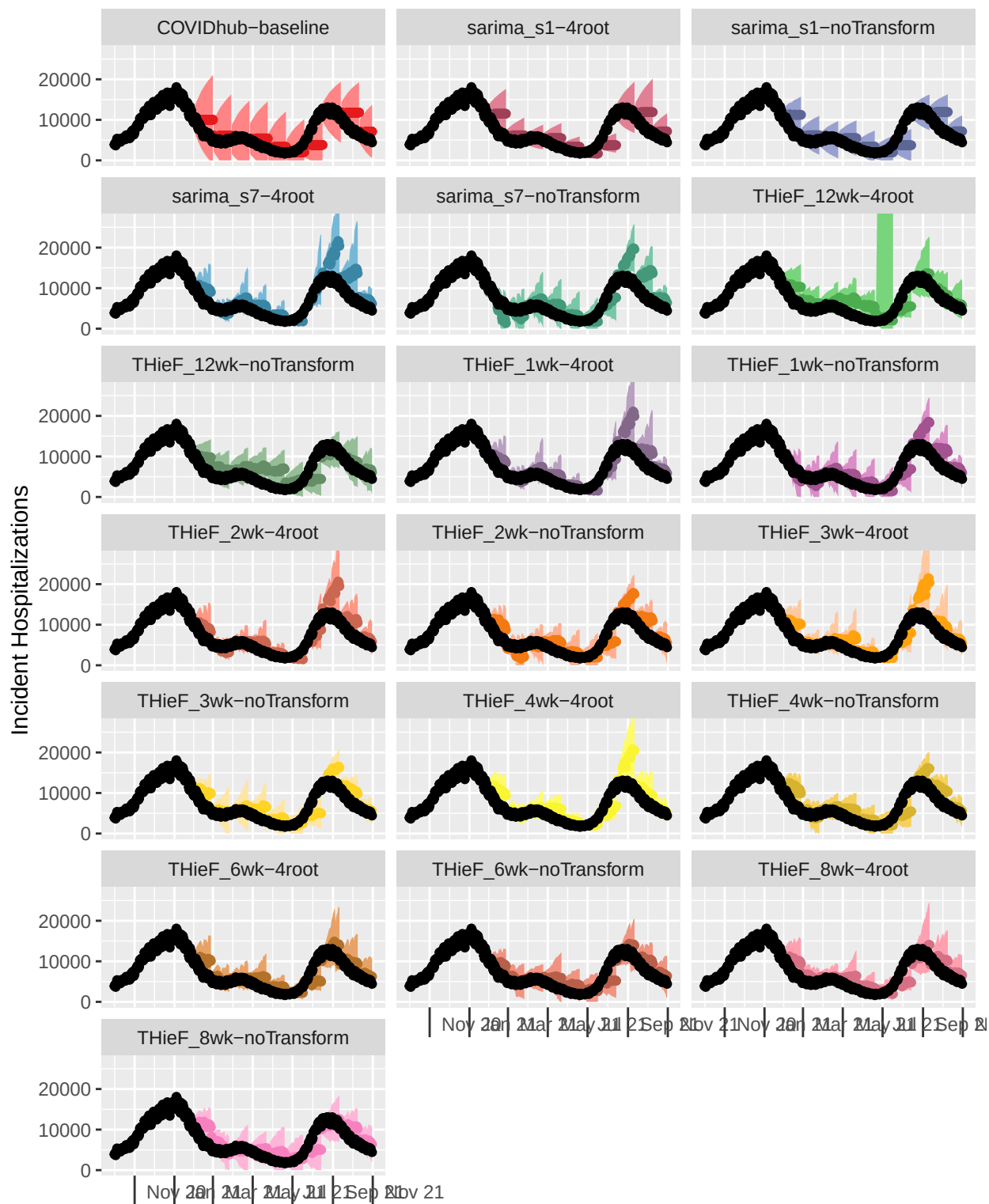
Plot Forecasts

We plot forecasts for three locations (US National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 10/31/21.

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): United States

Selected forecast date(s): 2021-01-29, 2021-02-26, 2021-03-27, 2021-04-24, 2021-05-21

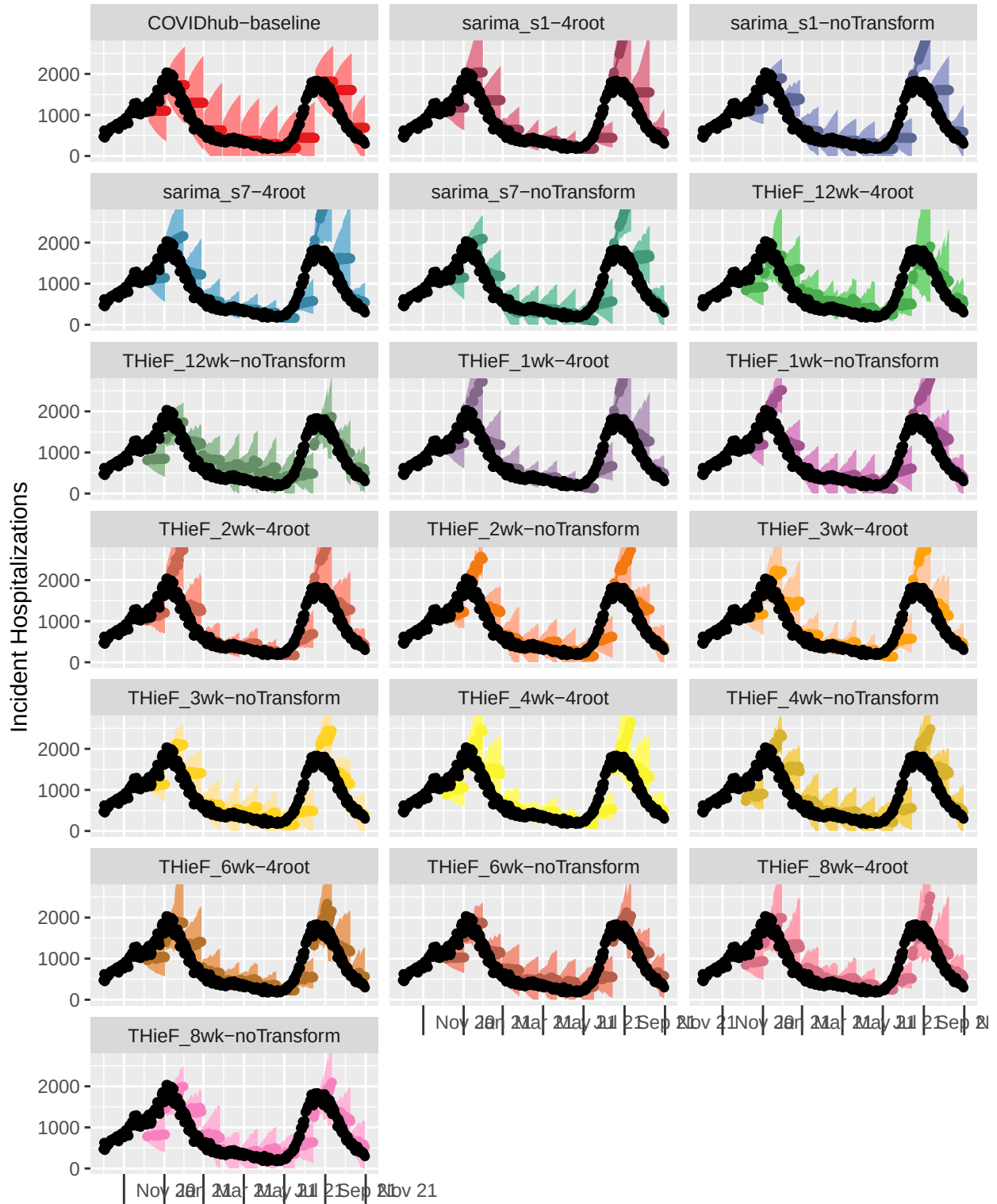


vk-4root, THieF_8wk-noTransform, THieF_12wk-4root, THieF_12wk-noTransform, COVIDhub-baseline (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Texas

Selected forecast date(s): 2020-12-05, 2021-01-02, 2021-01-29, 2021-02-26, 2021

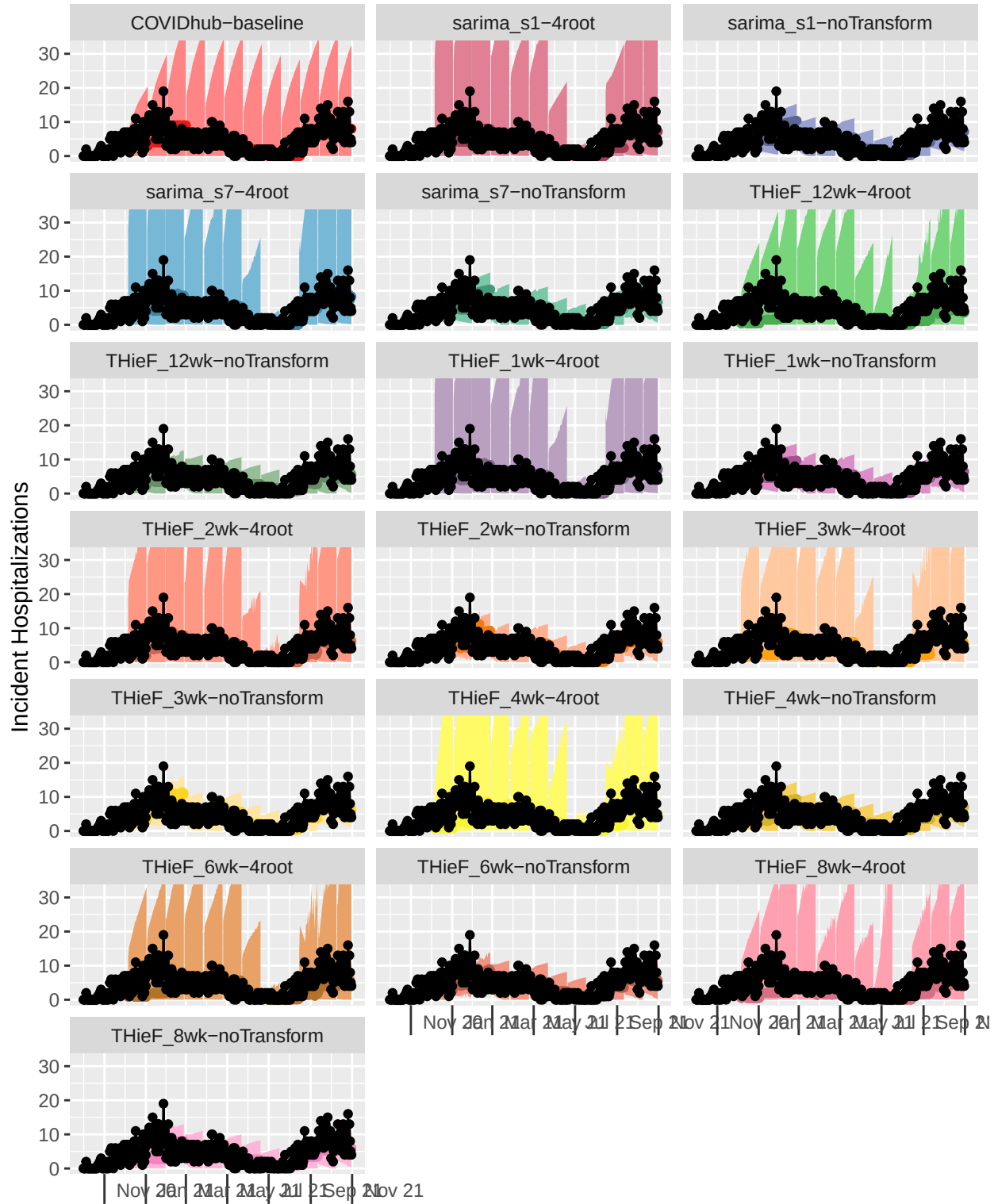


vk-4root, THieF_8wk-noTransform, THieF_12wk-4root, THieF_12wk-noTransform, COVIDhub-baseline (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Vermont

Selected forecast date(s): 2020-12-05, 2021-01-02, 2021-01-29, 2021-02-26, 2021-03-05



vk-4root, THieF_8wk-noTransform, THieF_12wk-4root, THieF_12wk-noTransform, COVIDhub-baseline (forecasts)

The plots don't show much difference between the 6 models (although this is partially due to the fact that the y-scales are free and are not consistent even for the same location). However, of note is the fact that the models that use a forth root transformation tend to have wider prediction intervals than the models without transformations. All of the models reasonably follow the trend of the truth lines for every location shown, although it seems like lower count locations may have slightly greater accuracy. The baseline seems to have intervals of fairly consistent widths while the models tend to vary in their interval widths, tending to show wider intervals during times of higher incidents or change points and narrow lower intervals otherwise. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

Overall metrics

model	wis	mae	cov_50	cov_95	rwis	rmae
THieF_6wk-noTransform	1071.3	1723.5	0.344	0.924	0.828	0.846
THieF_1wk-noTransform	1123.0	1766.7	0.450	0.886	0.868	0.868
THieF_6wk-4root	1133.3	1744.0	0.311	0.755	0.876	0.856
THieF_2wk-4root	1164.9	1867.0	0.228	0.860	0.900	0.917
THieF_1wk-4root	1175.4	1908.8	0.280	0.866	0.908	0.937
THieF_8wk-noTransform	1246.3	1948.7	0.371	0.856	0.963	0.957
THieF_2wk-noTransform	1251.3	1909.6	0.387	0.836	0.967	0.938
COVIDhub-baseline	1294.1	2036.2	0.622	0.986	1.000	1.000
THieF_8wk-4root	1297.7	1959.8	0.233	0.689	1.003	0.962
THieF_3wk-noTransform	1316.6	1953.5	0.368	0.788	1.017	0.959
THieF_3wk-4root	1329.9	2096.3	0.242	0.805	1.028	1.030
THieF_4wk-noTransform	1342.8	2004.8	0.331	0.776	1.038	0.985
THieF_4wk-4root	1348.7	2028.1	0.259	0.695	1.042	0.996
sarima_s7-noTransform	1401.9	2124.6	0.382	0.819	1.083	1.043
sarima_s1-noTransform	1419.3	2110.1	0.365	0.800	1.097	1.036
sarima_s1-4root	1482.8	2169.6	0.253	0.750	1.146	1.066
sarima_s7-4root	1537.6	2398.7	0.208	0.770	1.188	1.178
THieF_12wk-noTransform	1548.1	2378.9	0.301	0.764	1.196	1.168
THieF_12wk-4root	2334.4	2060.9	0.320	0.650	1.804	1.012

model	wis	mae	cov_50	cov_95	rwis	rmae
THieF_6wk-4root	31.653	48.108	0.456	0.908	0.882	0.963
THieF_6wk-noTransform	32.256	48.674	0.540	0.917	0.899	0.974
THieF_1wk-4root	32.678	50.063	0.480	0.911	0.911	1.002
THieF_2wk-4root	32.936	50.259	0.474	0.909	0.918	1.006
THieF_2wk-noTransform	33.258	48.027	0.574	0.900	0.927	0.961
THieF_1wk-noTransform	33.736	49.209	0.576	0.898	0.940	0.985
THieF_8wk-4root	33.844	51.142	0.428	0.887	0.943	1.023
THieF_4wk-4root	34.096	51.501	0.456	0.896	0.950	1.031

model	wis	mae	cov_50	cov_95	rwis	rmae
THieF_3wk-4root	34.622	52.708	0.460	0.899	0.965	1.055
THieF_8wk-noTransform	34.761	52.057	0.513	0.900	0.969	1.042
THieF_3wk-noTransform	34.960	52.077	0.535	0.896	0.974	1.042
THieF_4wk-noTransform	35.633	52.819	0.529	0.891	0.993	1.057
COVIDhub-baseline	35.877	49.975	0.826	0.983	1.000	1.000
sarima_s7-noTransform	36.292	51.906	0.567	0.886	1.012	1.039
sarima_s7-4root	36.346	54.549	0.465	0.898	1.013	1.092
sarima_s1-4root	36.913	55.430	0.455	0.887	1.029	1.109
THieF_12wk-noTransform	39.140	59.057	0.421	0.891	1.091	1.182
sarima_s1-noTransform	39.412	56.378	0.543	0.877	1.099	1.128
THieF_12wk-4root	84.321	54.493	0.408	0.864	2.350	1.090

Horizon week metrics

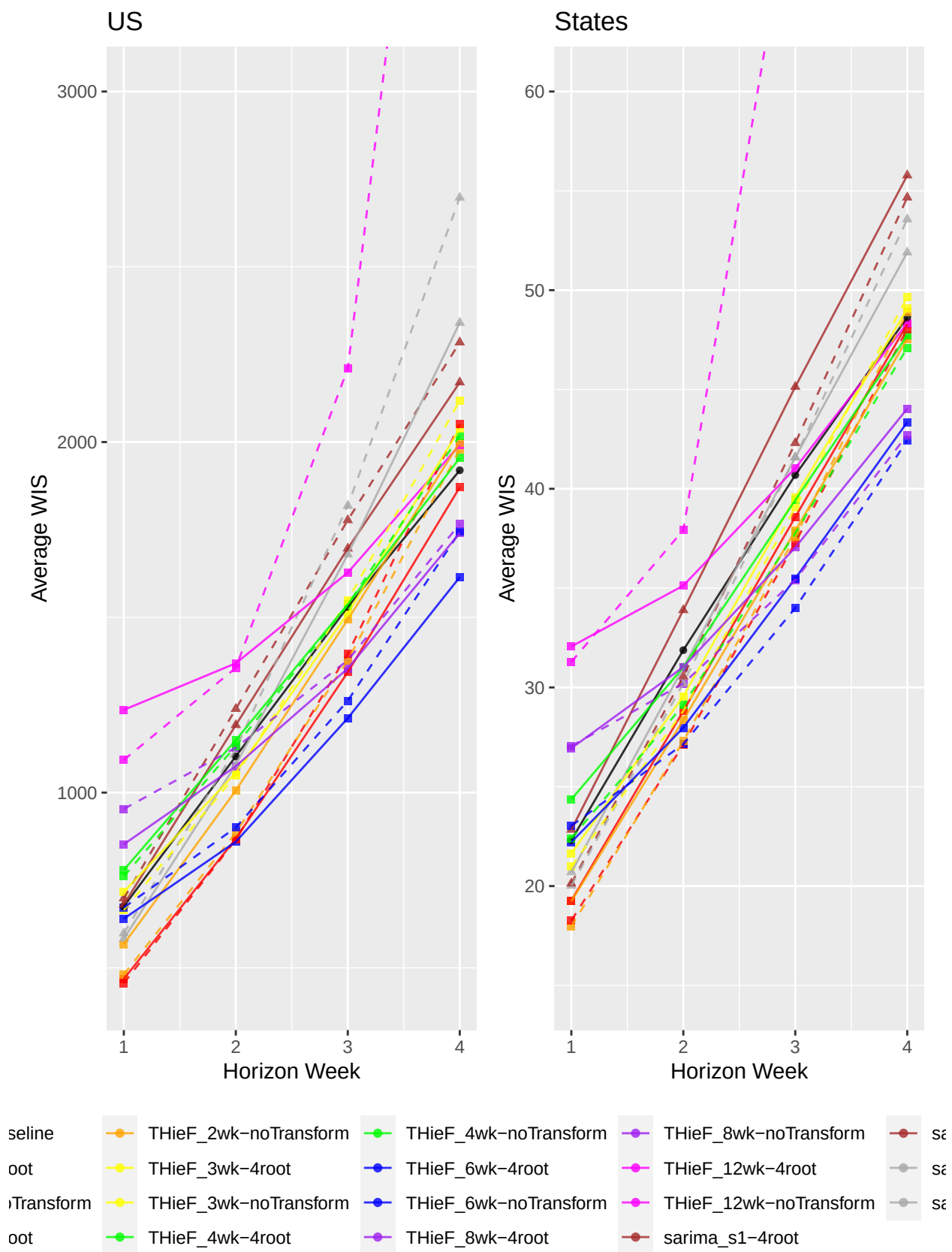
model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
THieF_1wk-4root	1	456.3	743.1	0.332	0.906	0.675	0.740
THieF_1wk-noTransform	1	468.5	713.5	0.465	0.892	0.694	0.710
THieF_2wk-4root	1	481.3	799.5	0.315	0.941	0.712	0.796
THieF_2wk-noTransform	1	567.5	877.1	0.455	0.902	0.840	0.873
sarima_s7-noTransform	1	583.2	872.9	0.395	0.818	0.863	0.869
sarima_s7-4root	1	598.7	942.2	0.238	0.780	0.886	0.938
THieF_6wk-noTransform	1	640.2	1047.2	0.318	0.937	0.948	1.042
THieF_3wk-4root	1	664.7	1057.3	0.210	0.794	0.984	1.052
THieF_6wk-4root	1	672.1	1065.7	0.308	0.766	0.995	1.061
COVIDhub-baseline	1	675.5	1004.6	0.706	1.000	1.000	1.000
sarima_s1-noTransform	1	680.8	1062.5	0.500	0.909	1.008	1.058
sarima_s1-4root	1	698.3	1104.2	0.360	0.885	1.034	1.099
THieF_3wk-noTransform	1	716.4	1100.0	0.346	0.846	1.060	1.095
THieF_4wk-4root	1	760.9	1183.9	0.259	0.755	1.126	1.179
THieF_4wk-noTransform	1	779.3	1201.5	0.322	0.822	1.154	1.196
THieF_8wk-noTransform	1	852.5	1346.1	0.304	0.825	1.262	1.340
THieF_8wk-4root	1	953.3	1454.7	0.157	0.622	1.411	1.448
THieF_12wk-4root	1	1093.1	1554.4	0.283	0.640	1.618	1.547
THieF_12wk-noTransform	1	1236.0	1854.2	0.241	0.717	1.830	1.846
THieF_6wk-noTransform	2	859.8	1401.0	0.373	0.953	0.779	0.816
THieF_1wk-4root	2	865.9	1406.0	0.290	0.871	0.785	0.819
THieF_1wk-noTransform	2	868.5	1339.1	0.419	0.885	0.787	0.780
THieF_2wk-4root	2	885.0	1444.3	0.247	0.882	0.802	0.841
THieF_6wk-4root	2	900.3	1412.5	0.344	0.803	0.816	0.823
THieF_2wk-noTransform	2	1006.0	1535.6	0.380	0.835	0.912	0.895
THieF_3wk-4root	2	1049.5	1660.2	0.251	0.810	0.951	0.967
THieF_3wk-noTransform	2	1056.7	1575.8	0.412	0.792	0.958	0.918
THieF_8wk-noTransform	2	1075.3	1729.9	0.391	0.885	0.975	1.008
sarima_s7-noTransform	2	1079.1	1628.4	0.398	0.828	0.978	0.949
COVIDhub-baseline	2	1103.4	1716.6	0.664	0.997	1.000	1.000
sarima_s7-4root	2	1125.0	1762.7	0.215	0.789	1.020	1.027
THieF_8wk-4root	2	1128.2	1741.6	0.247	0.717	1.022	1.015
THieF_4wk-4root	2	1133.2	1707.6	0.323	0.724	1.027	0.995
THieF_4wk-noTransform	2	1149.8	1701.7	0.373	0.796	1.042	0.991
sarima_s1-noTransform	2	1192.8	1816.5	0.358	0.835	1.081	1.058

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
sarima_s1-4root	2	1238.8	1871.1	0.229	0.756	1.123	1.090
THieF_12wk-4root	2	1355.8	1790.0	0.344	0.713	1.229	1.043
THieF_12wk-noTransform	2	1368.9	2149.0	0.262	0.792	1.241	1.252
THieF_6wk-noTransform	3	1212.2	1959.4	0.368	0.919	0.792	0.805
THieF_6wk-4root	3	1262.0	1942.3	0.342	0.746	0.825	0.798
THieF_1wk-noTransform	3	1343.5	2110.2	0.449	0.875	0.878	0.867
THieF_8wk-noTransform	3	1354.1	2109.5	0.423	0.875	0.885	0.867
THieF_8wk-4root	3	1377.6	2074.5	0.276	0.710	0.901	0.853
THieF_2wk-4root	3	1377.7	2207.4	0.184	0.835	0.901	0.907
THieF_1wk-4root	3	1395.9	2278.9	0.246	0.842	0.913	0.937
THieF_2wk-noTransform	3	1493.7	2269.1	0.364	0.809	0.976	0.933
THieF_3wk-noTransform	3	1511.2	2225.6	0.371	0.743	0.988	0.915
COVIDhub-baseline	3	1529.7	2433.1	0.581	0.978	1.000	1.000
THieF_4wk-4root	3	1535.8	2295.7	0.268	0.673	1.004	0.944
THieF_4wk-noTransform	3	1537.1	2272.6	0.357	0.750	1.005	0.934
THieF_3wk-4root	3	1549.1	2437.4	0.261	0.787	1.013	1.002
THieF_12wk-noTransform	3	1627.6	2518.8	0.342	0.798	1.064	1.035
sarima_s7-noTransform	3	1679.5	2542.7	0.375	0.820	1.098	1.045
sarima_s1-noTransform	3	1696.4	2507.9	0.301	0.743	1.109	1.031
sarima_s1-4root	3	1777.2	2573.7	0.206	0.676	1.162	1.058
sarima_s7-4root	3	1818.3	2837.9	0.195	0.776	1.189	1.166
THieF_12wk-4root	3	2209.7	2155.1	0.324	0.662	1.445	0.886
THieF_6wk-noTransform	4	1614.6	2551.0	0.317	0.883	0.841	0.829
THieF_8wk-noTransform	4	1740.6	2664.2	0.370	0.838	0.907	0.866
THieF_6wk-4root	4	1744.2	2621.4	0.249	0.702	0.909	0.852
THieF_8wk-4root	4	1766.0	2616.8	0.257	0.709	0.920	0.851
THieF_1wk-noTransform	4	1871.0	3000.8	0.468	0.891	0.975	0.975
COVIDhub-baseline	4	1919.5	3076.4	0.529	0.967	1.000	1.000
THieF_4wk-noTransform	4	1954.8	2916.1	0.272	0.732	1.018	0.948
THieF_2wk-4root	4	1978.8	3114.6	0.158	0.777	1.031	1.012
THieF_12wk-noTransform	4	1991.9	3043.7	0.366	0.751	1.038	0.989
THieF_2wk-noTransform	4	1998.9	3048.8	0.343	0.792	1.041	0.991
THieF_4wk-4root	4	2018.1	3002.0	0.181	0.623	1.051	0.976
THieF_3wk-noTransform	4	2038.4	2993.0	0.343	0.766	1.062	0.973
THieF_1wk-4root	4	2051.1	3316.3	0.249	0.842	1.069	1.078
THieF_3wk-4root	4	2118.0	3326.7	0.249	0.830	1.103	1.081
sarima_s1-noTransform	4	2170.3	3141.6	0.291	0.706	1.131	1.021
sarima_s1-4root	4	2284.3	3218.7	0.211	0.672	1.190	1.046
sarima_s7-noTransform	4	2340.3	3568.8	0.358	0.808	1.219	1.160
sarima_s7-4root	4	2697.4	4189.4	0.181	0.736	1.405	1.362
THieF_12wk-4root	4	4832.4	2796.2	0.332	0.581	2.518	0.909

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
THieF_2wk-4root	1	17.959	27.779	0.523	0.937	0.805	0.929
THieF_1wk-4root	1	18.266	27.836	0.524	0.933	0.819	0.931
THieF_2wk-noTransform	1	19.237	27.333	0.601	0.916	0.862	0.914
THieF_1wk-noTransform	1	19.268	27.524	0.598	0.912	0.864	0.920
sarima_s7-4root	1	20.022	30.107	0.505	0.920	0.897	1.006
sarima_s1-4root	1	20.133	30.394	0.505	0.924	0.902	1.016
sarima_s7-noTransform	1	20.699	29.235	0.583	0.902	0.928	0.977

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
THieF_3wk-4root	1	20.990	31.924	0.489	0.922	0.941	1.067
THieF_3wk-noTransform	1	21.647	31.781	0.551	0.906	0.970	1.062
THieF_6wk-noTransform	1	22.217	32.910	0.521	0.918	0.996	1.100
COVIDhub-baseline	1	22.310	29.915	0.816	0.981	1.000	1.000
THieF_4wk-4root	1	22.399	33.832	0.467	0.903	1.004	1.131
sarima_s1-noTransform	1	22.839	32.331	0.577	0.903	1.024	1.081
THieF_6wk-4root	1	23.034	34.513	0.447	0.898	1.032	1.154
THieF_4wk-noTransform	1	24.362	35.779	0.514	0.896	1.092	1.196
THieF_8wk-noTransform	1	26.941	39.108	0.485	0.887	1.208	1.307
THieF_8wk-4root	1	27.049	39.673	0.409	0.851	1.212	1.326
THieF_12wk-4root	1	31.284	44.609	0.384	0.822	1.402	1.491
THieF_12wk-noTransform	1	32.063	46.715	0.387	0.854	1.437	1.562
THieF_1wk-4root	2	27.118	42.031	0.491	0.924	0.851	0.972
THieF_6wk-4root	2	27.148	41.687	0.468	0.921	0.852	0.964
THieF_2wk-4root	2	27.297	42.323	0.479	0.919	0.856	0.979
THieF_6wk-noTransform	2	27.969	42.298	0.553	0.926	0.877	0.978
THieF_2wk-noTransform	2	28.360	40.791	0.585	0.902	0.890	0.944
THieF_3wk-4root	2	28.813	44.507	0.472	0.913	0.904	1.029
THieF_1wk-noTransform	2	28.815	41.786	0.586	0.903	0.904	0.967
THieF_4wk-4root	2	29.114	44.423	0.465	0.906	0.913	1.028
THieF_3wk-noTransform	2	29.556	43.815	0.555	0.901	0.927	1.013
THieF_8wk-4root	2	30.200	45.944	0.432	0.896	0.947	1.063
sarima_s7-4root	2	30.209	45.717	0.474	0.907	0.948	1.057
sarima_s1-4root	2	30.551	46.367	0.465	0.901	0.958	1.072
sarima_s7-noTransform	2	30.991	44.087	0.576	0.890	0.972	1.020
THieF_4wk-noTransform	2	31.021	46.196	0.536	0.899	0.973	1.069
THieF_8wk-noTransform	2	31.026	46.430	0.514	0.907	0.973	1.074
COVIDhub-baseline	2	31.875	43.233	0.839	0.986	1.000	1.000
sarima_s1-noTransform	2	33.894	48.541	0.559	0.884	1.063	1.123
THieF_12wk-noTransform	2	35.126	53.252	0.430	0.900	1.102	1.232
THieF_12wk-4root	2	37.923	49.396	0.417	0.873	1.190	1.143
THieF_6wk-4root	3	34.004	52.027	0.465	0.916	0.836	0.913
THieF_8wk-4root	3	35.416	54.220	0.437	0.905	0.870	0.952
THieF_6wk-noTransform	3	35.483	53.902	0.547	0.918	0.872	0.946
THieF_8wk-noTransform	3	37.044	55.968	0.524	0.906	0.910	0.983
THieF_1wk-4root	3	37.284	57.435	0.463	0.905	0.916	1.008
THieF_2wk-4root	3	37.587	57.458	0.460	0.902	0.924	1.009
THieF_4wk-4root	3	37.775	57.268	0.454	0.896	0.928	1.005
THieF_2wk-noTransform	3	37.887	55.006	0.566	0.895	0.931	0.966
THieF_1wk-noTransform	3	38.576	56.482	0.567	0.894	0.948	0.992
THieF_3wk-4root	3	39.049	59.719	0.448	0.888	0.960	1.048
THieF_4wk-noTransform	3	39.432	58.444	0.540	0.889	0.969	1.026
THieF_3wk-noTransform	3	39.569	58.925	0.527	0.891	0.972	1.035
COVIDhub-baseline	3	40.692	56.960	0.831	0.983	1.000	1.000
THieF_12wk-noTransform	3	41.035	62.631	0.434	0.905	1.008	1.100
sarima_s7-noTransform	3	41.576	59.557	0.561	0.882	1.022	1.046
sarima_s7-4root	3	41.584	62.634	0.451	0.892	1.022	1.100
sarima_s1-4root	3	42.305	63.714	0.438	0.875	1.040	1.119
sarima_s1-noTransform	3	45.135	64.745	0.530	0.868	1.109	1.137
THieF_12wk-4root	3	71.204	56.917	0.419	0.884	1.750	0.999
THieF_6wk-4root	4	42.426	64.206	0.445	0.897	0.872	0.920
THieF_8wk-4root	4	42.713	64.729	0.433	0.897	0.878	0.927

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
THieF_6wk-noTransform	4	43.355	65.585	0.537	0.907	0.891	0.940
THieF_8wk-noTransform	4	44.034	66.723	0.527	0.900	0.905	0.956
THieF_4wk-4root	4	47.095	70.483	0.436	0.879	0.968	1.010
THieF_2wk-noTransform	4	47.550	68.979	0.543	0.886	0.978	0.988
THieF_4wk-noTransform	4	47.717	70.859	0.527	0.881	0.981	1.015
THieF_1wk-4root	4	48.046	72.948	0.444	0.881	0.988	1.045
THieF_1wk-noTransform	4	48.284	71.043	0.552	0.883	0.993	1.018
THieF_12wk-noTransform	4	48.336	73.630	0.430	0.903	0.994	1.055
COVIDhub-baseline	4	48.633	69.792	0.820	0.982	1.000	1.000
THieF_2wk-4root	4	48.899	73.473	0.435	0.877	1.005	1.053
THieF_3wk-noTransform	4	49.069	73.786	0.507	0.887	1.009	1.057
THieF_3wk-4root	4	49.637	74.682	0.432	0.873	1.021	1.070
sarima_s7-noTransform	4	51.901	74.747	0.546	0.870	1.067	1.071
sarima_s7-4root	4	53.568	79.738	0.432	0.874	1.101	1.143
sarima_s1-4root	4	54.665	81.245	0.413	0.848	1.124	1.164
sarima_s1-noTransform	4	55.781	79.897	0.506	0.854	1.147	1.145
THieF_12wk-4root	4	196.873	67.049	0.410	0.877	4.048	0.961



When splitting the results so that US National is separate, we find that for US National the two models that use the base set hierarchy tended to perform the best, with improvements seen over a naive baseline in all metrics except for 95%PI Coverage. These two models are followed by the Topmost8_arima_noTrans model, which showed more comparable performance to the baseline. The Base_arima_noTrans model performed the best in terms of PI Coverage, followed by the Base_arima_4root and Topmost8_arima_noTrans models (with the latter being more consistent with coverage rates for all horizons). In terms of WIS and MAE, the Base_arima_4root model performed the best, followed by the Base_arima_noTrans model. Overall, for the US National level, it seems that there is a pattern that for coverage rates, the models without transformations perform better (and these tend to have smaller prediction intervals) while for WIS and MAE the fourth root transform models perform better; these patterns hold at all horizons. Further, there is a general order to model accuracy in terms of hierarchical structure used: the Base set performs the best, followed by the Topmost8 set, and finally the Multiple3 set.

For the state level, we see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models. (We also see shrunk WIS and MAE values, although this is likely partially due to the fact that the states have smaller scale values than the National level.) Interestingly, we see fewer improvements to MAE over the baseline but more substantial improvements in terms of coverage rates. It's a little harder to definitively say which models have the best coverage rates, as the results differ between the 50% and 95% intervals and the models are quite similar in coverage rates, especially for the 95% intervals. However, it seems that the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model.

Metrics by wave

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	THieF_1wk-4root	1	294.8	471.1	0.348	0.902	0.592	0.866
alpha	THieF_2wk-4root	1	303.5	486.9	0.357	0.911	0.609	0.895
alpha	THieF_1wk-noTransform	1	316.6	495.4	0.571	0.982	0.636	0.910
alpha	sarima_s1-4root	1	321.3	521.9	0.473	0.929	0.645	0.959
alpha	THieF_2wk-noTransform	1	326.4	493.2	0.661	1.000	0.655	0.906
alpha	sarima_s7-noTransform	1	353.5	568.5	0.518	0.964	0.710	1.044
alpha	sarima_s1-noTransform	1	363.5	526.0	0.786	1.000	0.730	0.966
alpha	sarima_s7-4root	1	392.2	611.4	0.196	0.750	0.787	1.123
alpha	THieF_3wk-noTransform	1	457.0	750.2	0.438	0.955	0.917	1.378
alpha	THieF_3wk-4root	1	457.5	719.8	0.188	0.768	0.918	1.323
alpha	THieF_4wk-4root	1	485.4	750.2	0.277	0.750	0.974	1.378
alpha	COVIDhub-baseline	1	498.2	544.2	0.973	1.000	1.000	1.000
alpha	THieF_4wk-noTransform	1	523.1	871.0	0.357	0.955	1.050	1.600
alpha	THieF_8wk-noTransform	1	545.1	923.2	0.393	1.000	1.094	1.696
alpha	THieF_6wk-4root	1	560.5	839.3	0.268	0.545	1.125	1.542
alpha	THieF_6wk-noTransform	1	627.6	1062.1	0.259	0.982	1.260	1.952
alpha	THieF_8wk-4root	1	728.6	1064.3	0.143	0.518	1.462	1.955
alpha	THieF_12wk-4root	1	1341.1	1775.3	0.134	0.455	2.692	3.262

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	THieF_12wk- noTransform	1	1584.5	2371.6	0.071	0.571	3.181	4.358
alpha	THieF_1wk-4root	2	519.7	834.7	0.339	0.893	0.704	0.927
alpha	THieF_2wk- noTransform	2	525.4	831.9	0.625	1.000	0.711	0.924
alpha	THieF_2wk-4root	2	530.8	858.4	0.321	0.911	0.719	0.953
alpha	sarima_s1-4root	2	553.1	890.2	0.286	0.866	0.749	0.989
alpha	sarima_s1- noTransform	2	554.3	885.6	0.616	1.000	0.750	0.984
alpha	THieF_1wk- noTransform	2	575.8	927.0	0.527	1.000	0.779	1.030
alpha	sarima_s7- noTransform	2	623.3	1003.9	0.598	0.982	0.844	1.115
alpha	THieF_3wk- noTransform	2	652.4	1060.2	0.473	1.000	0.883	1.178
alpha	THieF_4wk-4root	2	653.3	1005.1	0.304	0.777	0.884	1.116
alpha	sarima_s7-4root	2	659.6	1040.2	0.196	0.830	0.893	1.155
alpha	THieF_8wk- noTransform	2	670.7	1103.6	0.616	1.000	0.908	1.226
alpha	THieF_3wk-4root	2	675.1	1058.9	0.295	0.777	0.914	1.176
alpha	THieF_4wk- noTransform	2	694.8	1149.7	0.402	1.000	0.941	1.277
alpha	THieF_6wk-4root	2	702.7	1080.0	0.250	0.705	0.951	1.200
alpha	COVIDhub-baseline	2	738.7	900.3	0.982	1.000	1.000	1.000
alpha	THieF_6wk- noTransform	2	773.2	1298.1	0.375	1.000	1.047	1.442
alpha	THieF_8wk-4root	2	810.3	1224.9	0.277	0.616	1.097	1.361
alpha	THieF_12wk- noTransform	2	1561.3	2532.8	0.098	0.741	2.114	2.813
alpha	THieF_12wk-4root	2	1630.7	1791.7	0.268	0.554	2.207	1.990
alpha	THieF_2wk- noTransform	3	798.5	1303.2	0.554	1.000	0.818	0.979
alpha	sarima_s1- noTransform	3	804.1	1312.1	0.473	0.973	0.824	0.986
alpha	THieF_1wk-4root	3	840.8	1347.1	0.241	0.830	0.862	1.012
alpha	THieF_2wk-4root	3	859.6	1378.4	0.188	0.839	0.881	1.035
alpha	sarima_s1-4root	3	874.6	1320.0	0.250	0.750	0.896	0.991
alpha	THieF_1wk- noTransform	3	906.2	1511.5	0.562	1.000	0.929	1.135
alpha	THieF_8wk- noTransform	3	909.1	1469.1	0.491	1.000	0.932	1.103
alpha	THieF_4wk- noTransform	3	935.2	1556.1	0.366	0.991	0.958	1.169
alpha	THieF_4wk-4root	3	935.4	1412.8	0.295	0.688	0.959	1.061
alpha	sarima_s7- noTransform	3	951.7	1517.6	0.562	1.000	0.975	1.140
alpha	COVIDhub-baseline	3	975.7	1331.3	0.920	1.000	1.000	1.000
alpha	THieF_3wk- noTransform	3	999.0	1659.5	0.384	0.955	1.024	1.246
alpha	THieF_6wk- noTransform	3	1004.2	1716.1	0.348	1.000	1.029	1.289
alpha	THieF_6wk-4root	3	1012.9	1552.3	0.205	0.634	1.038	1.166

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	sarima_s7-4root	3	1013.1	1587.5	0.214	0.804	1.038	1.192
alpha	THieF_8wk-4root	3	1053.6	1610.9	0.214	0.607	1.080	1.210
alpha	THieF_3wk-4root	3	1076.9	1669.5	0.188	0.768	1.104	1.254
alpha	THieF_12wk-noTransform	3	1769.7	2885.7	0.179	0.812	1.814	2.168
alpha	THieF_12wk-4root	3	3325.1	2161.3	0.179	0.491	3.408	1.623
alpha	THieF_2wk-noTransform	4	1107.7	1838.2	0.536	1.000	0.877	0.993
alpha	sarima_s1-noTransform	4	1150.3	1830.6	0.420	0.929	0.911	0.989
alpha	THieF_8wk-noTransform	4	1202.6	1900.7	0.438	0.973	0.952	1.027
alpha	THieF_6wk-noTransform	4	1262.4	2147.4	0.312	0.964	0.999	1.160
alpha	COVIDhub-baseline	4	1263.2	1851.0	0.857	1.000	1.000	1.000
alpha	THieF_2wk-4root	4	1264.0	1996.3	0.170	0.759	1.001	1.079
alpha	THieF_1wk-4root	4	1275.7	2009.1	0.232	0.812	1.010	1.085
alpha	sarima_s1-4root	4	1289.7	1840.6	0.223	0.723	1.021	0.994
alpha	THieF_1wk-noTransform	4	1318.0	2254.6	0.518	1.000	1.043	1.218
alpha	THieF_4wk-noTransform	4	1343.4	2220.5	0.268	0.929	1.064	1.200
alpha	sarima_s7-noTransform	4	1360.3	2189.6	0.527	1.000	1.077	1.183
alpha	THieF_4wk-4root	4	1402.1	2048.1	0.170	0.616	1.110	1.106
alpha	THieF_8wk-4root	4	1429.3	2129.4	0.188	0.652	1.132	1.150
alpha	THieF_6wk-4root	4	1461.7	2138.9	0.161	0.598	1.157	1.156
alpha	sarima_s7-4root	4	1471.2	2307.3	0.188	0.759	1.165	1.246
alpha	THieF_3wk-noTransform	4	1486.9	2464.2	0.295	0.938	1.177	1.331
alpha	THieF_3wk-4root	4	1638.3	2543.9	0.170	0.795	1.297	1.374
alpha	THieF_12wk-noTransform	4	2139.2	3471.8	0.179	0.768	1.694	1.876
alpha	THieF_12wk-4root	4	8990.8	2925.8	0.179	0.375	7.118	1.581
delta	THieF_2wk-4root	1	553.9	947.1	0.234	0.973	0.720	0.722
delta	THieF_1wk-4root	1	578.8	951.2	0.216	0.892	0.753	0.725
delta	THieF_1wk-noTransform	1	600.5	951.7	0.216	0.802	0.781	0.726
delta	THieF_2wk-noTransform	1	616.0	1004.2	0.288	0.874	0.801	0.766
delta	THieF_3wk-4root	1	677.7	1081.3	0.270	0.829	0.881	0.825
delta	THieF_6wk-4root	1	685.5	1112.9	0.360	0.928	0.892	0.849
delta	THieF_3wk-noTransform	1	740.8	1151.4	0.315	0.820	0.963	0.878
delta	THieF_6wk-noTransform	1	743.7	1176.4	0.288	0.856	0.967	0.897
delta	THieF_12wk-4root	1	752.8	1115.1	0.523	0.865	0.979	0.850
delta	THieF_4wk-noTransform	1	764.3	1181.6	0.324	0.784	0.994	0.901
delta	COVIDhub-baseline	1	768.8	1311.2	0.450	1.000	1.000	1.000
delta	sarima_s7-4root	1	786.2	1237.0	0.144	0.730	1.023	0.943

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
delta	sarima_s1- noTransform	1	822.7	1336.0	0.297	0.874	1.070	1.019
delta	sarima_s1-4root	1	825.3	1310.6	0.315	0.892	1.073	1.000
delta	sarima_s7- noTransform	1	825.8	1226.2	0.144	0.631	1.074	0.935
delta	THieF_4wk-4root	1	859.1	1341.9	0.306	0.793	1.117	1.023
delta	THieF_8wk- noTransform	1	873.0	1381.6	0.306	0.829	1.135	1.054
delta	THieF_12wk- noTransform	1	1045.7	1482.8	0.450	0.757	1.360	1.131
delta	THieF_8wk-4root	1	1083.0	1658.5	0.216	0.721	1.409	1.265
delta	THieF_6wk-4root	2	996.3	1529.2	0.471	0.837	0.719	0.643
delta	THieF_6wk- noTransform	2	1048.0	1715.9	0.240	0.904	0.756	0.722
delta	THieF_12wk-4root	2	1060.4	1538.0	0.577	0.846	0.765	0.647
delta	THieF_2wk-4root	2	1149.1	1872.5	0.183	0.846	0.829	0.788
delta	THieF_1wk- noTransform	2	1173.4	1881.3	0.144	0.760	0.847	0.791
delta	THieF_8wk- noTransform	2	1193.1	1950.2	0.269	0.865	0.861	0.820
delta	THieF_1wk-4root	2	1197.0	1957.6	0.192	0.837	0.864	0.823
delta	THieF_2wk- noTransform	2	1252.1	1928.3	0.183	0.683	0.904	0.811
delta	THieF_3wk-4root	2	1273.5	2066.2	0.221	0.865	0.919	0.869
delta	THieF_3wk- noTransform	2	1279.3	1906.4	0.308	0.615	0.923	0.802
delta	THieF_4wk- noTransform	2	1316.6	2010.3	0.269	0.663	0.950	0.846
delta	THieF_12wk- noTransform	2	1343.6	1925.9	0.481	0.769	0.970	0.810
delta	THieF_8wk-4root	2	1357.0	2057.2	0.288	0.760	0.979	0.865
delta	COVIDhub-baseline	2	1385.6	2377.6	0.327	0.990	1.000	1.000
delta	THieF_4wk-4root	2	1469.5	2255.9	0.279	0.702	1.061	0.949
delta	sarima_s1- noTransform	2	1581.6	2425.5	0.144	0.712	1.141	1.020
delta	sarima_s1-4root	2	1597.4	2397.0	0.192	0.731	1.153	1.008
delta	sarima_s7- noTransform	2	1612.9	2428.3	0.038	0.625	1.164	1.021
delta	sarima_s7-4root	2	1625.7	2554.2	0.115	0.712	1.173	1.074
delta	THieF_12wk-4root	3	1322.7	1931.5	0.577	0.804	0.657	0.566
delta	THieF_6wk-4root	3	1412.2	2142.1	0.526	0.845	0.701	0.628
delta	THieF_6wk- noTransform	3	1492.5	2367.4	0.309	0.825	0.741	0.694
delta	THieF_8wk- noTransform	3	1519.0	2453.4	0.340	0.866	0.754	0.719
delta	THieF_12wk- noTransform	3	1620.3	2295.2	0.536	0.722	0.804	0.673
delta	THieF_8wk-4root	3	1621.4	2391.5	0.330	0.794	0.805	0.701
delta	THieF_2wk-4root	3	1865.2	2962.6	0.227	0.825	0.926	0.868
delta	THieF_1wk- noTransform	3	1869.2	2988.8	0.186	0.722	0.928	0.876

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
delta	THieF_4wk-noTransform	3	1909.4	2806.1	0.247	0.526	0.948	0.823
delta	THieF_2wk-noTransform	3	1921.1	2854.6	0.258	0.608	0.954	0.837
delta	THieF_3wk-noTransform	3	1928.9	2777.7	0.227	0.474	0.958	0.814
delta	THieF_3wk-4root	3	1981.2	3229.3	0.361	0.897	0.984	0.947
delta	COVIDhub-baseline	3	2014.3	3411.3	0.216	0.938	1.000	1.000
delta	THieF_1wk-4root	3	2073.5	3422.0	0.196	0.856	1.029	1.003
delta	THieF_4wk-4root	3	2081.4	3156.4	0.247	0.680	1.033	0.925
delta	sarima_s1-noTransform	3	2364.2	3459.1	0.103	0.577	1.174	1.014
delta	sarima_s1-4root	3	2370.7	3429.1	0.144	0.691	1.177	1.005
delta	sarima_s7-noTransform	3	2603.9	3891.1	0.062	0.567	1.293	1.141
delta	sarima_s7-4root	3	2900.4	4517.3	0.113	0.742	1.440	1.324
delta	THieF_12wk-4root	4	1667.0	2527.1	0.533	0.844	0.647	0.584
delta	THieF_12wk-noTransform	4	1954.4	2784.1	0.522	0.722	0.759	0.644
delta	THieF_6wk-4root	4	1986.2	3007.4	0.344	0.844	0.771	0.696
delta	THieF_8wk-noTransform	4	2012.7	3190.0	0.278	0.811	0.782	0.738
delta	THieF_6wk-noTransform	4	2058.4	3152.0	0.211	0.778	0.799	0.729
delta	THieF_8wk-4root	4	2079.2	3067.3	0.278	0.800	0.807	0.709
delta	THieF_4wk-noTransform	4	2443.0	3555.4	0.200	0.533	0.949	0.822
delta	THieF_3wk-noTransform	4	2562.4	3714.4	0.189	0.544	0.995	0.859
delta	COVIDhub-baseline	4	2574.9	4323.7	0.156	0.900	1.000	1.000
delta	THieF_1wk-noTransform	4	2638.8	4183.7	0.278	0.756	1.025	0.968
delta	THieF_3wk-4root	4	2688.3	4407.1	0.333	0.967	1.044	1.019
delta	THieF_2wk-noTransform	4	2716.1	3950.3	0.211	0.533	1.055	0.914
delta	THieF_4wk-4root	4	2739.3	4173.8	0.189	0.622	1.064	0.965
delta	THieF_2wk-4root	4	2879.3	4516.4	0.167	0.722	1.118	1.045
delta	sarima_s1-4root	4	3051.6	4343.4	0.133	0.711	1.185	1.005
delta	sarima_s1-noTransform	4	3059.1	4374.8	0.089	0.522	1.188	1.012
delta	THieF_1wk-4root	4	3232.4	5301.8	0.278	0.844	1.255	1.226
delta	sarima_s7-noTransform	4	3746.3	5558.1	0.089	0.511	1.455	1.285
delta	sarima_s7-4root	4	4703.2	7253.4	0.156	0.678	1.827	1.678
winter21	THieF_6wk-noTransform	1	480.4	792.8	0.476	1.000	0.592	0.632
winter21	THieF_1wk-noTransform	1	505.9	681.6	0.714	0.889	0.624	0.543
winter21	THieF_1wk-4root	1	527.6	860.1	0.508	0.937	0.650	0.685
winter21	sarima_s7-noTransform	1	564.3	791.5	0.619	0.889	0.696	0.631
winter21	sarima_s7-4root	1	635.4	1010.9	0.476	0.921	0.783	0.806

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
winter21	THieF_2wk-4root	1	669.6	1094.9	0.381	0.937	0.825	0.873
winter21	COVIDhub-baseline	1	811.3	1254.9	0.686	1.000	1.000	1.000
winter21	THieF_6wk-4root	1	846.9	1385.2	0.286	0.873	1.044	1.104
winter21	THieF_2wk- noTransform	1	910.8	1335.9	0.381	0.778	1.123	1.065
winter21	THieF_12wk- noTransform	1	951.5	1588.7	0.175	0.905	1.173	1.266
winter21	sarima_s1- noTransform	1	994.6	1534.4	0.349	0.810	1.226	1.223
winter21	THieF_3wk-4root	1	1010.2	1615.0	0.143	0.778	1.245	1.287
winter21	THieF_4wk-4root	1	1077.8	1676.5	0.143	0.698	1.328	1.336
winter21	THieF_8wk-4root	1	1124.1	1789.7	0.079	0.635	1.386	1.426
winter21	THieF_3wk- noTransform	1	1134.5	1631.5	0.238	0.698	1.398	1.300
winter21	sarima_s1-4root	1	1144.6	1775.7	0.238	0.794	1.411	1.415
winter21	THieF_12wk-4root	1	1251.9	1935.7	0.127	0.571	1.543	1.543
winter21	THieF_4wk- noTransform	1	1261.2	1824.1	0.254	0.651	1.555	1.454
winter21	THieF_8wk- noTransform	1	1362.8	2035.3	0.143	0.508	1.680	1.622
winter21	THieF_6wk- noTransform	2	703.3	1064.1	0.587	0.952	0.555	0.521
winter21	THieF_1wk- noTransform	2	885.4	1176.7	0.683	0.889	0.698	0.577
winter21	THieF_1wk-4root	2	934.8	1511.0	0.365	0.889	0.737	0.740
winter21	sarima_s7- noTransform	2	1008.3	1418.3	0.635	0.889	0.795	0.695
winter21	THieF_12wk- noTransform	2	1068.7	1835.1	0.190	0.921	0.843	0.899
winter21	THieF_2wk-4root	2	1078.6	1779.2	0.222	0.889	0.851	0.872
winter21	THieF_6wk-4root	2	1092.9	1811.0	0.302	0.921	0.862	0.887
winter21	sarima_s7-4root	2	1125.7	1740.5	0.413	0.841	0.888	0.853
winter21	COVIDhub-baseline	2	1267.7	2040.6	0.657	1.000	1.000	1.000
winter21	THieF_8wk-4root	2	1315.6	2138.9	0.127	0.825	1.038	1.048
winter21	THieF_3wk-4root	2	1345.3	2059.0	0.222	0.778	1.061	1.009
winter21	THieF_12wk-4root	2	1354.7	2202.9	0.095	0.778	1.069	1.080
winter21	THieF_3wk- noTransform	2	1407.9	1946.9	0.476	0.714	1.111	0.954
winter21	THieF_4wk-4root	2	1431.2	2051.2	0.429	0.667	1.129	1.005
winter21	THieF_2wk- noTransform	2	1454.2	2138.1	0.270	0.794	1.147	1.048
winter21	THieF_8wk- noTransform	2	1599.9	2479.8	0.190	0.714	1.262	1.215
winter21	THieF_4wk- noTransform	2	1683.3	2173.6	0.492	0.651	1.328	1.065
winter21	sarima_s1- noTransform	2	1686.2	2466.1	0.254	0.746	1.330	1.209
winter21	sarima_s1-4root	2	1865.8	2747.0	0.190	0.603	1.472	1.346
winter21	THieF_6wk- noTransform	3	1150.3	1763.6	0.492	0.921	0.659	0.621
winter21	THieF_1wk- noTransform	3	1311.4	1821.7	0.651	0.889	0.752	0.641

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
winter21	THieF_1wk-4root	3	1339.3	2175.3	0.333	0.841	0.768	0.766
winter21	THieF_12wk- noTransform	3	1386.5	2210.8	0.333	0.889	0.795	0.778
winter21	THieF_6wk-4root	3	1473.5	2328.0	0.302	0.794	0.845	0.820
winter21	THieF_2wk-4root	3	1548.2	2518.4	0.111	0.841	0.888	0.887
winter21	sarima_s7- noTransform	3	1550.1	2289.2	0.524	0.889	0.889	0.806
winter21	THieF_8wk-4root	3	1578.2	2410.7	0.302	0.762	0.905	0.849
winter21	sarima_s7-4root	3	1583.8	2475.1	0.286	0.778	0.908	0.871
winter21	THieF_12wk-4root	3	1592.4	2488.5	0.190	0.746	0.913	0.876
winter21	THieF_3wk-4root	3	1723.0	2583.5	0.238	0.651	0.988	0.910
winter21	COVIDhub-baseline	3	1744.4	2840.4	0.543	1.000	1.000	1.000
winter21	THieF_4wk-4root	3	1763.1	2540.0	0.254	0.635	1.011	0.894
winter21	THieF_3wk- noTransform	3	1778.4	2382.0	0.571	0.778	1.020	0.839
winter21	THieF_8wk- noTransform	3	1891.3	2718.5	0.429	0.667	1.084	0.957
winter21	THieF_4wk- noTransform	3	2033.8	2725.0	0.508	0.667	1.166	0.959
winter21	THieF_2wk- noTransform	3	2071.4	3084.8	0.190	0.778	1.187	1.086
winter21	sarima_s1- noTransform	3	2254.4	3169.1	0.302	0.587	1.292	1.116
winter21	sarima_s1-4root	3	2467.8	3485.6	0.222	0.524	1.415	1.227
winter21	THieF_6wk- noTransform	4	1606.7	2409.8	0.476	0.889	0.755	0.702
winter21	THieF_1wk-4root	4	1742.2	2803.8	0.238	0.889	0.819	0.817
winter21	THieF_1wk- noTransform	4	1757.3	2637.7	0.651	0.889	0.826	0.768
winter21	THieF_12wk- noTransform	4	1783.6	2653.6	0.476	0.762	0.839	0.773
winter21	THieF_6wk-4root	4	1900.7	2927.7	0.270	0.683	0.894	0.853
winter21	THieF_8wk-4root	4	1917.2	2839.9	0.349	0.683	0.901	0.827
winter21	THieF_12wk-4root	4	1961.6	2950.2	0.317	0.571	0.922	0.859
winter21	THieF_2wk-4root	4	1963.2	3100.0	0.127	0.889	0.923	0.903
winter21	sarima_s7-4root	4	2011.9	3158.3	0.206	0.778	0.946	0.920
winter21	sarima_s7- noTransform	4	2074.0	3178.7	0.444	0.889	0.975	0.926
winter21	THieF_4wk-4root	4	2082.9	3023.7	0.190	0.635	0.979	0.881
winter21	COVIDhub-baseline	4	2126.8	3433.3	0.486	1.000	1.000	1.000
winter21	THieF_3wk-4root	4	2156.2	3174.9	0.270	0.698	1.014	0.925
winter21	THieF_3wk- noTransform	4	2270.2	2902.4	0.651	0.778	1.067	0.845
winter21	THieF_8wk- noTransform	4	2308.4	3270.5	0.381	0.635	1.085	0.953
winter21	THieF_4wk- noTransform	4	2344.1	3239.5	0.381	0.667	1.102	0.944
winter21	THieF_2wk- noTransform	4	2558.5	3913.1	0.190	0.794	1.203	1.140
winter21	sarima_s1- noTransform	4	2714.2	3710.5	0.349	0.571	1.276	1.081
winter21	sarima_s1-4root	4	2956.2	4062.1	0.302	0.524	1.390	1.183

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	THieF_2wk-4root	1	8.843	13.759	0.527	0.941	0.548	0.815
alpha	THieF_1wk-4root	1	9.177	14.301	0.515	0.934	0.569	0.847
alpha	sarima_s1-4root	1	9.545	14.894	0.523	0.938	0.592	0.882
alpha	sarima_s7-4root	1	9.604	15.011	0.505	0.930	0.595	0.889
alpha	THieF_3wk-4root	1	10.049	15.818	0.496	0.930	0.623	0.937
alpha	THieF_1wk- noTransform	1	10.812	14.710	0.771	0.983	0.670	0.871
alpha	THieF_2wk- noTransform	1	10.893	14.592	0.773	0.986	0.675	0.864
alpha	sarima_s7- noTransform	1	11.009	14.817	0.765	0.980	0.683	0.878
alpha	THieF_4wk-4root	1	11.300	17.621	0.462	0.907	0.701	1.044
alpha	sarima_s1- noTransform	1	11.755	15.743	0.773	0.981	0.729	0.933
alpha	THieF_3wk- noTransform	1	12.310	18.759	0.705	0.985	0.763	1.111
alpha	THieF_6wk-4root	1	12.408	19.609	0.422	0.890	0.769	1.162
alpha	THieF_4wk- noTransform	1	13.722	21.687	0.646	0.986	0.851	1.285
alpha	THieF_8wk- noTransform	1	14.565	22.493	0.636	0.979	0.903	1.333
alpha	THieF_8wk-4root	1	14.931	23.120	0.380	0.834	0.926	1.370
alpha	THieF_6wk- noTransform	1	15.423	25.717	0.539	0.987	0.956	1.524
alpha	COVIDhub-baseline	1	16.131	16.879	0.948	0.994	1.000	1.000
alpha	THieF_12wk-4root	1	22.950	32.795	0.324	0.771	1.423	1.943
alpha	THieF_12wk- noTransform	1	27.292	44.228	0.300	0.893	1.692	2.620
alpha	THieF_2wk-4root	2	13.446	20.858	0.492	0.931	0.604	0.885
alpha	THieF_1wk-4root	2	13.592	21.198	0.511	0.934	0.610	0.899
alpha	sarima_s7-4root	2	14.048	21.702	0.508	0.927	0.631	0.920
alpha	sarima_s1-4root	2	14.060	21.658	0.511	0.929	0.631	0.918
alpha	THieF_3wk-4root	2	14.568	22.744	0.480	0.927	0.654	0.965
alpha	THieF_4wk-4root	2	15.271	23.796	0.467	0.925	0.686	1.009
alpha	THieF_1wk- noTransform	2	15.693	22.077	0.774	0.981	0.705	0.936
alpha	THieF_2wk- noTransform	2	15.735	21.547	0.769	0.985	0.707	0.914
alpha	sarima_s7- noTransform	2	15.739	21.478	0.779	0.977	0.707	0.911
alpha	THieF_6wk-4root	2	15.797	25.077	0.437	0.924	0.709	1.063
alpha	sarima_s1- noTransform	2	16.440	22.253	0.776	0.979	0.738	0.944
alpha	THieF_3wk- noTransform	2	16.789	25.414	0.725	0.984	0.754	1.078
alpha	THieF_8wk-4root	2	17.582	27.663	0.412	0.894	0.790	1.173
alpha	THieF_8wk- noTransform	2	17.699	26.440	0.689	0.987	0.795	1.121
alpha	THieF_4wk- noTransform	2	17.776	27.795	0.686	0.986	0.798	1.179
alpha	THieF_6wk- noTransform	2	19.425	31.827	0.606	0.988	0.872	1.350

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	COVIDhub-baseline	2	22.267	23.580	0.957	0.995	1.000	1.000
alpha	THieF_12wk- noTransform	2	29.300	49.292	0.369	0.963	1.316	2.090
alpha	THieF_12wk-4root	2	38.024	36.365	0.369	0.840	1.708	1.542
alpha	THieF_1wk-4root	3	19.977	30.494	0.482	0.922	0.714	0.958
alpha	THieF_2wk-4root	3	20.102	30.443	0.461	0.922	0.719	0.957
alpha	sarima_s1-4root	3	20.243	30.187	0.486	0.910	0.724	0.949
alpha	sarima_s7-4root	3	20.256	30.722	0.488	0.919	0.724	0.966
alpha	THieF_4wk-4root	3	21.407	32.617	0.452	0.921	0.765	1.025
alpha	THieF_6wk-4root	3	21.435	33.299	0.428	0.921	0.766	1.047
alpha	THieF_3wk-4root	3	21.517	32.763	0.444	0.900	0.769	1.030
alpha	sarima_s7- noTransform	3	21.721	30.684	0.758	0.974	0.777	0.964
alpha	THieF_1wk- noTransform	3	21.931	31.984	0.748	0.977	0.784	1.005
alpha	sarima_s1- noTransform	3	21.958	30.430	0.738	0.974	0.785	0.956
alpha	THieF_2wk- noTransform	3	22.078	31.458	0.733	0.980	0.789	0.989
alpha	THieF_8wk- noTransform	3	22.587	34.065	0.689	0.984	0.808	1.071
alpha	THieF_8wk-4root	3	22.821	35.565	0.402	0.914	0.816	1.118
alpha	THieF_4wk- noTransform	3	23.146	35.988	0.695	0.978	0.828	1.131
alpha	THieF_3wk- noTransform	3	23.577	36.486	0.677	0.977	0.843	1.147
alpha	THieF_6wk- noTransform	3	24.893	40.185	0.618	0.983	0.890	1.263
alpha	COVIDhub-baseline	3	27.968	31.818	0.948	0.993	1.000	1.000
alpha	THieF_12wk- noTransform	3	34.831	58.662	0.384	0.969	1.245	1.844
alpha	THieF_12wk-4root	3	126.715	44.356	0.365	0.859	4.531	1.394
alpha	sarima_s1-4root	4	28.477	40.814	0.452	0.874	0.830	0.972
alpha	THieF_1wk-4root	4	28.491	42.006	0.449	0.895	0.830	1.000
alpha	THieF_8wk- noTransform	4	28.501	43.156	0.682	0.975	0.830	1.027
alpha	sarima_s7-4root	4	28.528	41.710	0.454	0.893	0.831	0.993
alpha	THieF_2wk-4root	4	28.815	42.263	0.425	0.888	0.839	1.006
alpha	sarima_s1- noTransform	4	28.835	40.658	0.703	0.963	0.840	0.968
alpha	sarima_s7- noTransform	4	29.059	41.948	0.729	0.963	0.847	0.999
alpha	THieF_6wk-4root	4	29.188	43.894	0.403	0.899	0.850	1.045
alpha	THieF_2wk- noTransform	4	29.331	42.151	0.701	0.968	0.854	1.004
alpha	THieF_1wk- noTransform	4	29.463	43.731	0.723	0.967	0.858	1.041
alpha	THieF_4wk-4root	4	29.935	43.877	0.416	0.892	0.872	1.045
alpha	THieF_8wk-4root	4	30.176	45.634	0.396	0.905	0.879	1.086
alpha	THieF_4wk- noTransform	4	30.303	46.903	0.662	0.964	0.883	1.117

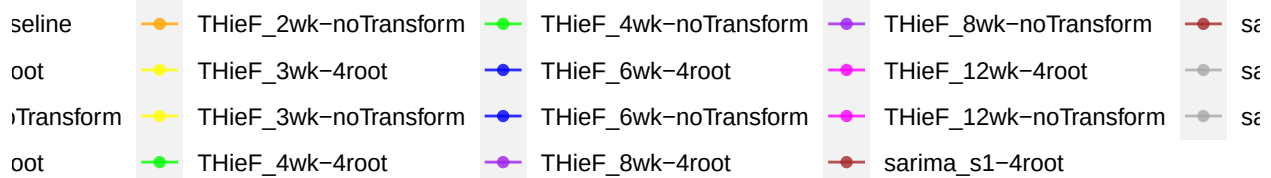
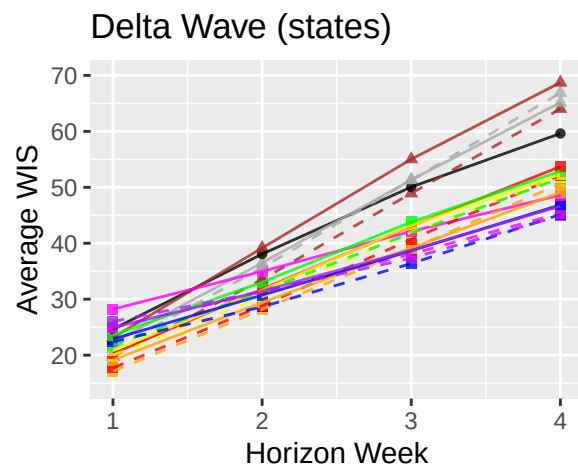
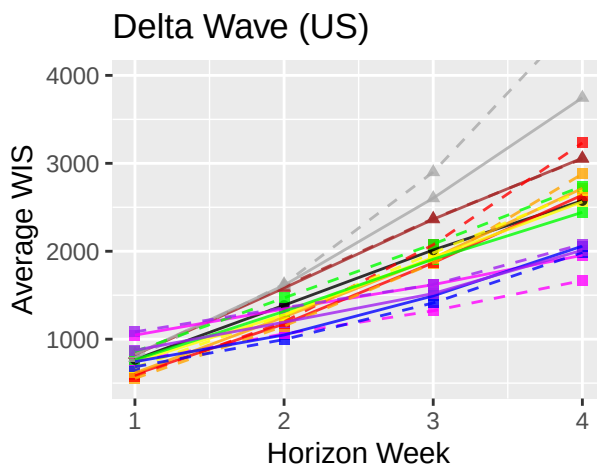
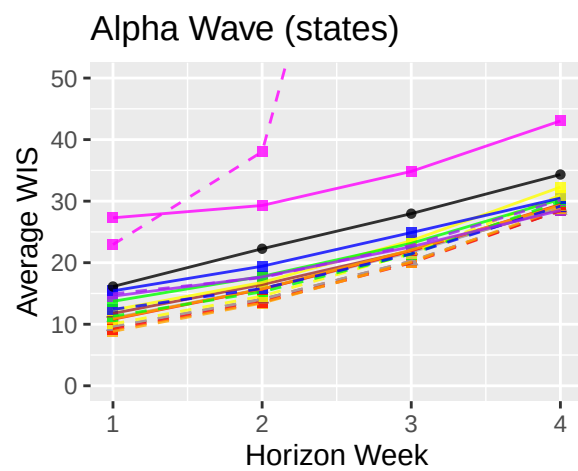
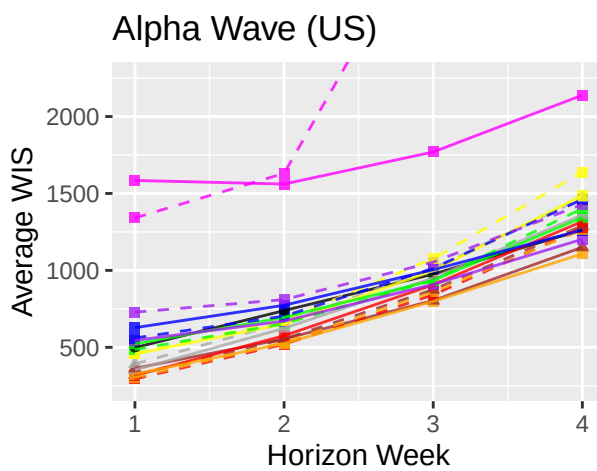
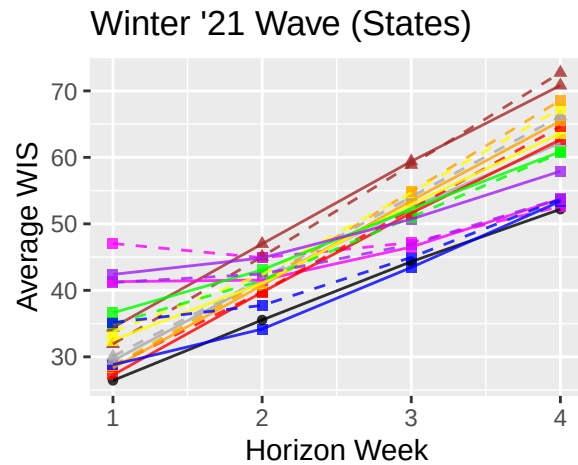
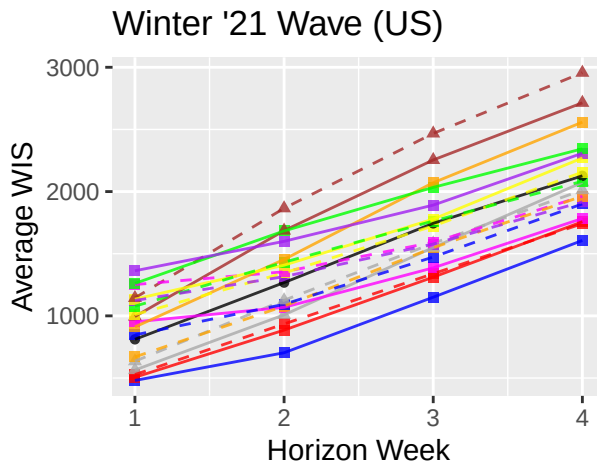
wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	THieF_6wk- noTransform	4	30.499	47.889	0.637	0.973	0.889	1.140
alpha	THieF_3wk-4root	4	30.963	45.743	0.401	0.880	0.902	1.089
alpha	THieF_3wk- noTransform	4	32.277	51.096	0.623	0.969	0.940	1.217
alpha	COVIDhub-baseline	4	34.326	42.002	0.930	0.991	1.000	1.000
alpha	THieF_12wk- noTransform	4	43.033	71.829	0.373	0.964	1.254	1.710
alpha	THieF_12wk-4root	4	482.735	56.833	0.348	0.847	14.063	1.353
delta	THieF_2wk-4root	1	17.117	26.578	0.539	0.945	0.695	0.735
delta	THieF_1wk-4root	1	17.713	27.689	0.532	0.943	0.719	0.766
delta	THieF_2wk- noTransform	1	19.117	27.527	0.531	0.897	0.776	0.761
delta	sarima_s1-4root	1	19.732	30.568	0.503	0.930	0.801	0.846
delta	THieF_3wk-4root	1	20.002	30.483	0.510	0.930	0.812	0.843
delta	THieF_1wk- noTransform	1	20.269	29.576	0.504	0.886	0.823	0.818
delta	THieF_3wk- noTransform	1	20.860	30.399	0.511	0.884	0.847	0.841
delta	sarima_s7-4root	1	21.094	32.455	0.498	0.920	0.856	0.898
delta	THieF_4wk-4root	1	21.785	32.633	0.507	0.916	0.884	0.903
delta	sarima_s7- noTransform	1	22.277	32.170	0.479	0.867	0.904	0.890
delta	THieF_6wk-4root	1	22.383	33.376	0.489	0.917	0.909	0.923
delta	THieF_6wk- noTransform	1	22.876	33.136	0.500	0.883	0.929	0.917
delta	sarima_s1- noTransform	1	23.144	33.894	0.478	0.874	0.939	0.938
delta	THieF_4wk- noTransform	1	23.584	33.893	0.495	0.872	0.957	0.937
delta	COVIDhub-baseline	1	24.637	36.152	0.716	0.972	1.000	1.000
delta	THieF_8wk- noTransform	1	24.923	35.858	0.485	0.871	1.012	0.992
delta	THieF_12wk-4root	1	24.931	36.375	0.494	0.906	1.012	1.006
delta	THieF_8wk-4root	1	26.010	38.121	0.461	0.890	1.056	1.054
delta	THieF_12wk- noTransform	1	28.228	39.426	0.476	0.839	1.146	1.091
delta	THieF_2wk-4root	2	28.151	43.242	0.500	0.921	0.740	0.762
delta	THieF_6wk-4root	2	28.642	42.877	0.512	0.923	0.752	0.755
delta	THieF_1wk-4root	2	28.832	44.944	0.490	0.922	0.757	0.792
delta	THieF_2wk- noTransform	2	29.422	42.924	0.508	0.876	0.773	0.756
delta	THieF_3wk-4root	2	30.439	46.335	0.501	0.906	0.800	0.816
delta	THieF_6wk- noTransform	2	30.731	45.131	0.518	0.889	0.807	0.795
delta	THieF_12wk-4root	2	31.321	45.905	0.499	0.914	0.823	0.809
delta	THieF_8wk-4root	2	31.400	46.685	0.485	0.908	0.825	0.823
delta	THieF_4wk-4root	2	31.436	46.948	0.493	0.897	0.826	0.827
delta	THieF_8wk- noTransform	2	31.456	46.398	0.497	0.877	0.826	0.818
delta	THieF_3wk- noTransform	2	31.608	45.968	0.494	0.870	0.830	0.810

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
delta	THieF_1wk- noTransform	2	31.783	46.745	0.483	0.868	0.835	0.824
delta	THieF_4wk- noTransform	2	33.040	48.561	0.481	0.864	0.868	0.856
delta	sarima_s1-4root	2	33.524	51.293	0.448	0.893	0.881	0.904
delta	THieF_12wk- noTransform	2	34.973	49.279	0.497	0.854	0.919	0.868
delta	sarima_s7-4root	2	35.617	53.896	0.450	0.894	0.936	0.950
delta	sarima_s7- noTransform	2	36.488	52.231	0.457	0.844	0.959	0.920
delta	COVIDhub-baseline	2	38.067	56.755	0.723	0.972	1.000	1.000
delta	sarima_s1- noTransform	2	39.151	56.855	0.436	0.841	1.028	1.002
delta	THieF_6wk-4root	3	36.372	53.911	0.508	0.914	0.727	0.714
delta	THieF_8wk-4root	3	37.330	55.151	0.505	0.910	0.746	0.730
delta	THieF_12wk-4root	3	38.068	55.041	0.495	0.911	0.761	0.729
delta	THieF_6wk- noTransform	3	38.661	57.250	0.511	0.879	0.773	0.758
delta	THieF_8wk- noTransform	3	38.775	57.642	0.500	0.881	0.775	0.763
delta	THieF_2wk-4root	3	38.991	59.030	0.495	0.903	0.779	0.782
delta	THieF_2wk- noTransform	3	39.153	57.544	0.503	0.870	0.782	0.762
delta	THieF_1wk-4root	3	40.707	62.441	0.470	0.895	0.813	0.827
delta	THieF_4wk-4root	3	41.824	62.384	0.478	0.884	0.836	0.826
delta	THieF_3wk-4root	3	41.997	63.577	0.483	0.884	0.839	0.842
delta	THieF_12wk- noTransform	3	42.134	58.490	0.493	0.859	0.842	0.774
delta	THieF_1wk- noTransform	3	43.072	63.570	0.460	0.857	0.861	0.842
delta	THieF_3wk- noTransform	3	43.159	62.801	0.472	0.857	0.862	0.832
delta	THieF_4wk- noTransform	3	43.804	64.424	0.461	0.853	0.875	0.853
delta	sarima_s1-4root	3	48.925	73.609	0.421	0.864	0.978	0.975
delta	COVIDhub-baseline	3	50.044	75.526	0.705	0.965	1.000	1.000
delta	sarima_s7-4root	3	51.297	75.911	0.431	0.866	1.025	1.005
delta	sarima_s7- noTransform	3	51.326	72.880	0.440	0.825	1.026	0.965
delta	sarima_s1- noTransform	3	55.021	79.067	0.405	0.817	1.099	1.047
delta	THieF_8wk-4root	4	44.907	66.666	0.487	0.902	0.753	0.733
delta	THieF_6wk-4root	4	45.143	67.184	0.482	0.901	0.757	0.738
delta	THieF_12wk-4root	4	45.372	64.617	0.474	0.902	0.761	0.710
delta	THieF_8wk- noTransform	4	46.640	70.382	0.483	0.875	0.782	0.773
delta	THieF_6wk- noTransform	4	46.858	70.722	0.472	0.874	0.786	0.777
delta	THieF_12wk- noTransform	4	48.614	67.409	0.486	0.860	0.815	0.741
delta	THieF_2wk- noTransform	4	49.075	72.591	0.476	0.868	0.823	0.798

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
delta	THieF_2wk-4root	4	50.648	76.480	0.465	0.886	0.850	0.840
delta	THieF_4wk-4root	4	51.612	77.311	0.463	0.877	0.866	0.850
delta	THieF_3wk-4root	4	51.841	78.166	0.478	0.877	0.870	0.859
delta	THieF_1wk-4root	4	52.152	79.195	0.455	0.875	0.875	0.870
delta	THieF_3wk- noTransform	4	52.315	77.297	0.456	0.856	0.877	0.849
delta	THieF_4wk- noTransform	4	52.897	78.704	0.440	0.844	0.887	0.865
delta	THieF_1wk- noTransform	4	53.715	79.799	0.446	0.843	0.901	0.877
delta	COVIDhub-baseline	4	59.620	90.998	0.692	0.965	1.000	1.000
delta	sarima_s1-4root	4	64.026	95.362	0.400	0.845	1.074	1.048
delta	sarima_s7- noTransform	4	65.158	92.398	0.427	0.811	1.093	1.015
delta	sarima_s7-4root	4	66.871	97.487	0.426	0.853	1.122	1.071
delta	sarima_s1- noTransform	4	68.721	98.267	0.378	0.797	1.153	1.080
winter21	COVIDhub-baseline	1	26.445	37.213	0.783	0.977	1.000	1.000
winter21	THieF_1wk- noTransform	1	27.270	39.076	0.514	0.864	1.031	1.050
winter21	THieF_2wk- noTransform	1	28.323	40.803	0.493	0.860	1.071	1.096
winter21	THieF_1wk-4root	1	28.616	42.525	0.525	0.921	1.082	1.143
winter21	THieF_2wk-4root	1	28.651	44.121	0.500	0.925	1.083	1.186
winter21	THieF_6wk- noTransform	1	28.802	40.390	0.525	0.881	1.089	1.085
winter21	sarima_s7- noTransform	1	29.406	41.562	0.501	0.855	1.112	1.117
winter21	sarima_s7-4root	1	30.056	43.792	0.511	0.909	1.137	1.177
winter21	sarima_s1-4root	1	31.931	46.848	0.486	0.903	1.207	1.259
winter21	THieF_3wk- noTransform	1	32.516	47.244	0.428	0.845	1.230	1.270
winter21	THieF_3wk-4root	1	33.797	50.763	0.458	0.904	1.278	1.364
winter21	sarima_s1- noTransform	1	34.411	48.460	0.473	0.849	1.301	1.302
winter21	THieF_4wk-4root	1	34.972	52.524	0.430	0.884	1.322	1.411
winter21	THieF_6wk-4root	1	35.142	51.733	0.429	0.886	1.329	1.390
winter21	THieF_4wk- noTransform	1	36.620	52.933	0.391	0.826	1.385	1.422
winter21	THieF_8wk-4root	1	41.175	59.110	0.384	0.829	1.557	1.588
winter21	THieF_12wk- noTransform	1	41.303	57.208	0.385	0.829	1.562	1.537
winter21	THieF_8wk- noTransform	1	42.393	60.436	0.324	0.805	1.603	1.624
winter21	THieF_12wk-4root	1	47.050	66.133	0.331	0.788	1.779	1.777
winter21	THieF_6wk- noTransform	2	34.175	50.498	0.535	0.901	0.962	1.014
winter21	COVIDhub-baseline	2	35.542	49.816	0.836	0.990	1.000	1.000
winter21	THieF_6wk-4root	2	37.730	58.242	0.453	0.918	1.062	1.169
winter21	THieF_1wk- noTransform	2	39.715	57.621	0.495	0.856	1.117	1.157
winter21	THieF_1wk-4root	2	39.799	61.270	0.469	0.914	1.120	1.230

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
winter21	THieF_2wk- noTransform	2	40.771	59.161	0.469	0.842	1.147	1.188
winter21	THieF_3wk- noTransform	2	41.058	61.256	0.437	0.844	1.155	1.230
winter21	THieF_2wk-4root	2	41.250	64.381	0.443	0.905	1.161	1.292
winter21	sarima_s7- noTransform	2	41.464	59.614	0.487	0.846	1.167	1.197
winter21	THieF_4wk-4root	2	41.483	63.855	0.432	0.897	1.167	1.282
winter21	THieF_12wk- noTransform	2	41.544	61.768	0.425	0.882	1.169	1.240
winter21	sarima_s7-4root	2	41.752	62.716	0.465	0.899	1.175	1.259
winter21	THieF_3wk-4root	2	42.361	65.908	0.432	0.905	1.192	1.323
winter21	THieF_8wk-4root	2	42.455	64.775	0.396	0.886	1.195	1.300
winter21	THieF_4wk- noTransform	2	43.072	63.412	0.434	0.842	1.212	1.273
winter21	THieF_8wk- noTransform	2	44.874	67.923	0.344	0.855	1.263	1.363
winter21	THieF_12wk-4root	2	44.902	67.133	0.380	0.864	1.263	1.348
winter21	sarima_s1-4root	2	45.062	67.606	0.434	0.879	1.268	1.357
winter21	sarima_s1- noTransform	2	46.990	67.836	0.460	0.827	1.322	1.362
winter21	THieF_6wk- noTransform	3	43.440	65.034	0.511	0.890	0.980	1.016
winter21	COVIDhub-baseline	3	44.312	64.019	0.840	0.992	1.000	1.000
winter21	THieF_6wk-4root	3	44.956	70.110	0.459	0.913	1.015	1.095
winter21	THieF_12wk- noTransform	3	46.517	71.337	0.425	0.886	1.050	1.114
winter21	THieF_8wk-4root	3	46.882	73.246	0.401	0.892	1.058	1.144
winter21	THieF_12wk-4root	3	47.184	72.416	0.397	0.883	1.065	1.131
winter21	THieF_8wk- noTransform	3	50.706	77.684	0.371	0.850	1.144	1.213
winter21	THieF_4wk-4root	3	51.002	78.241	0.431	0.881	1.151	1.222
winter21	THieF_1wk- noTransform	3	51.619	75.172	0.488	0.845	1.165	1.174
winter21	THieF_1wk-4root	3	52.188	80.982	0.435	0.898	1.178	1.265
winter21	THieF_4wk- noTransform	3	52.223	76.131	0.460	0.832	1.179	1.189
winter21	sarima_s7- noTransform	3	52.425	76.252	0.479	0.843	1.183	1.191
winter21	THieF_3wk- noTransform	3	52.885	78.854	0.425	0.834	1.193	1.232
winter21	THieF_2wk- noTransform	3	53.500	77.562	0.453	0.832	1.207	1.212
winter21	sarima_s7-4root	3	54.054	82.641	0.431	0.890	1.220	1.291
winter21	THieF_3wk-4root	3	54.705	84.514	0.416	0.879	1.235	1.320
winter21	THieF_2wk-4root	3	54.851	84.772	0.420	0.879	1.238	1.324
winter21	sarima_s1-4root	3	58.883	89.083	0.403	0.849	1.329	1.391
winter21	sarima_s1- noTransform	3	59.403	86.210	0.441	0.810	1.341	1.347
winter21	COVIDhub-baseline	4	52.196	76.859	0.839	0.990	1.000	1.000
winter21	THieF_12wk-4root	4	52.631	80.627	0.406	0.881	1.008	1.049

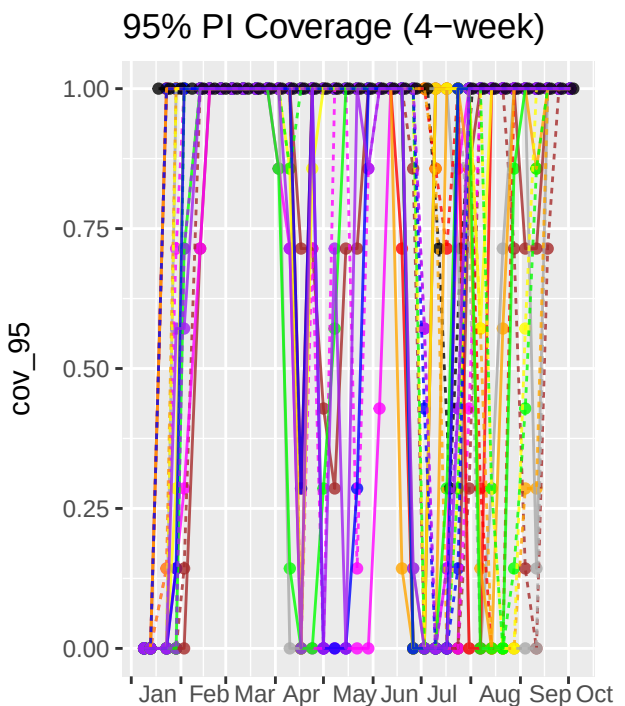
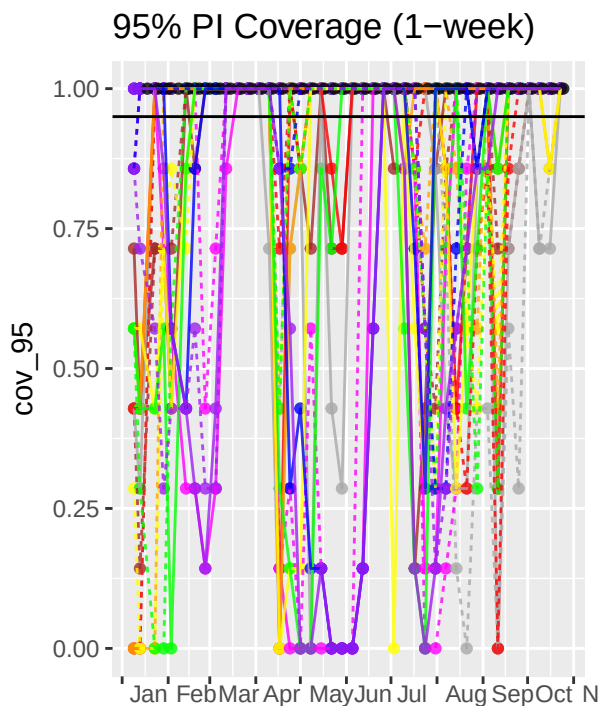
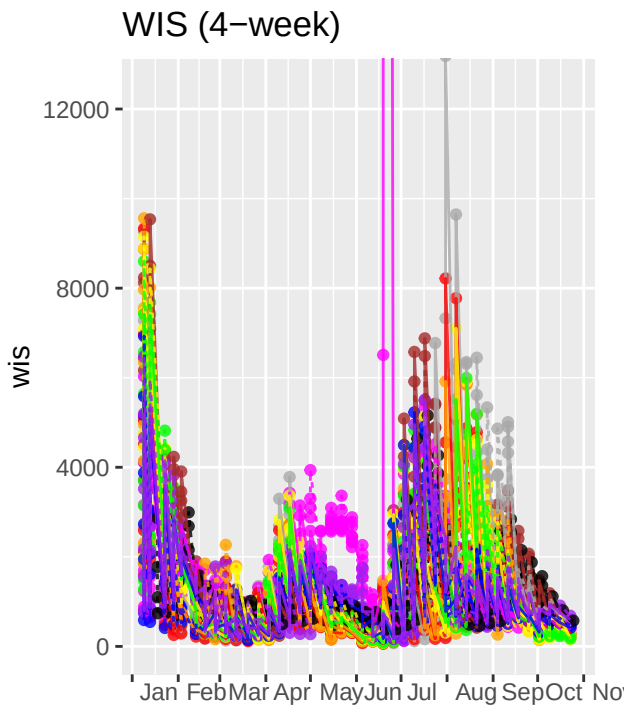
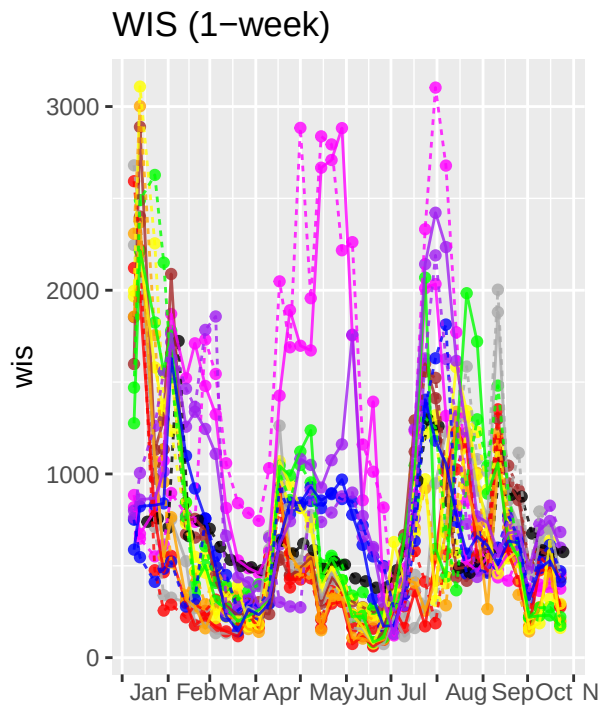
wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
winter21	THieF_6wk- noTransform	4	53.396	79.066	0.500	0.871	1.023	1.029
winter21	THieF_6wk-4root	4	53.722	82.813	0.451	0.890	1.029	1.077
winter21	THieF_12wk- noTransform	4	53.730	82.243	0.432	0.883	1.029	1.070
winter21	THieF_8wk-4root	4	53.816	83.148	0.415	0.882	1.031	1.082
winter21	THieF_8wk- noTransform	4	57.911	88.095	0.410	0.848	1.109	1.146
winter21	THieF_4wk-4root	4	60.668	91.715	0.430	0.869	1.162	1.193
winter21	THieF_4wk- noTransform	4	60.851	88.155	0.475	0.831	1.166	1.147
winter21	sarima_s7- noTransform	4	62.193	91.008	0.477	0.834	1.192	1.184
winter21	THieF_1wk- noTransform	4	62.660	90.964	0.482	0.836	1.200	1.184
winter21	THieF_3wk- noTransform	4	63.610	94.374	0.436	0.832	1.219	1.228
winter21	THieF_1wk-4root	4	64.630	99.459	0.425	0.873	1.238	1.294
winter21	THieF_2wk- noTransform	4	65.471	93.901	0.446	0.818	1.254	1.222
winter21	sarima_s7-4root	4	66.169	101.509	0.412	0.874	1.268	1.321
winter21	THieF_3wk-4root	4	67.320	102.009	0.416	0.861	1.290	1.327
winter21	THieF_2wk-4root	4	68.582	103.751	0.414	0.858	1.314	1.350
winter21	sarima_s1- noTransform	4	70.817	102.300	0.432	0.798	1.357	1.331
winter21	sarima_s1-4root	4	72.729	109.494	0.384	0.825	1.393	1.425



For US National during the alpha wave, the Base_arima_noTrans model performed the best for coverage rates and second best for WIS and MAE at every horizon. The Base_arima_4root model performed best in terms of WIS and MAE but was third for coverage rate while the Topmost8_arima_noTrans model had the second best coverage rates and third best WIS and MAE. During the alpha wave, these models performed the best and tended to have more consistent values for their metrics. The winter21 wave saw noticeably worse values for all metrics at every horizon. Generally, the two models that use the base set hierarchy tended to perform the best for all metrics, although the coverage rates varied a lot by horizon, with the Topmost8_arima_4root and Multiple3_arima_noTrans achieving closer to nominal coverage rates for the 95% interval than even the Base models at horizons greater than 1 week ahead. These results are pretty consistent with improvements over the base-line, which ranged from modest to significant for the best performing models during the alpha wave, especially for WIS and 50% PI coverage. The winter21 wave saw fairly comparable or worse performance compared to the baseline for WIS and MAE. The 50% coverage was better than the baseline for all models, the greatest improvements for the best models, but the 95% coverage could be better or worse for the best models.

For states during both waves, we again see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models, compared to those of the US National level when broken into waves. During the alpha wave, the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model. During the winter21 wave, the Base_arima_4root model is again generally the best for all metrics at all horizon weeks but sometimes loses out in terms of 95% coverage at longer horizons to other models. Notably during this wave, there are much greater similarities in coverage rates and certain models have better coverage rates for the 50% interval while others have better coverage reads for the 95% one. Similar patterns to the ones described above for the US national level when compared to the baseline were observed.

By forecast date (week)



THieF_2wk-noTransform

THieF_3wk-4root

THieF_3wk-noTransform

THieF_4wk-4root

● THieF_4wk-noTransform

THieF_6wk-4root

THieF_6wk-noTransform

THieF_8wk-4root

THieF_8wk-noTransform

THieF_12wk-4root

THieF_12wk-noTransform

—●— sarima_s1-4root

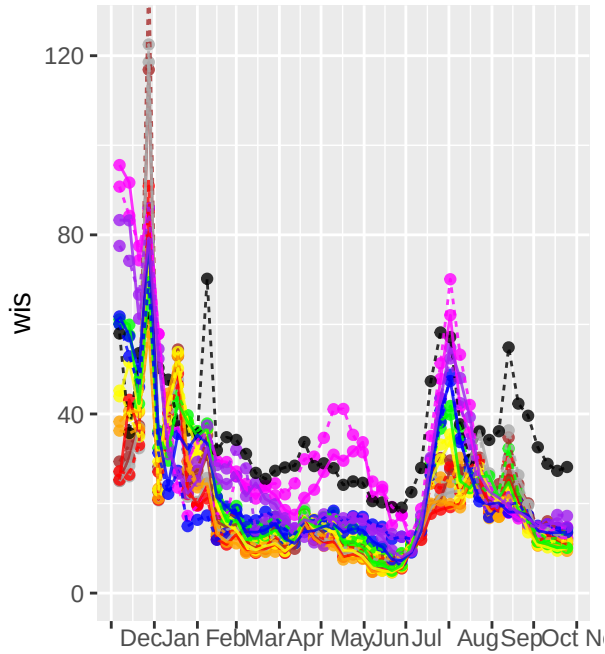
—●— sarima_s1-noTrar

 sarima_s7-4root

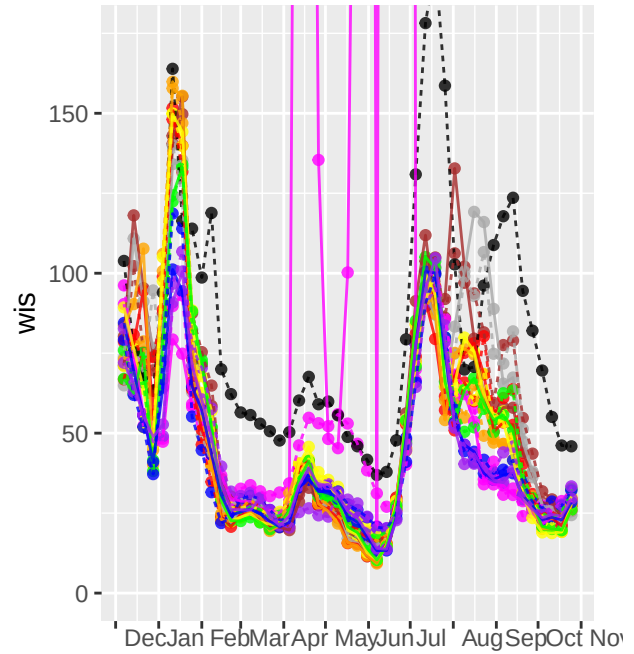
☐ sarima_s7-noTrar

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## 'summarise()' has grouped output by 'forecast_date', 'model'. You can override
## using the '.groups' argument.
## 'summarise()' has grouped output by 'forecast_date', 'model'. You can override
## using the '.groups' argument.
```

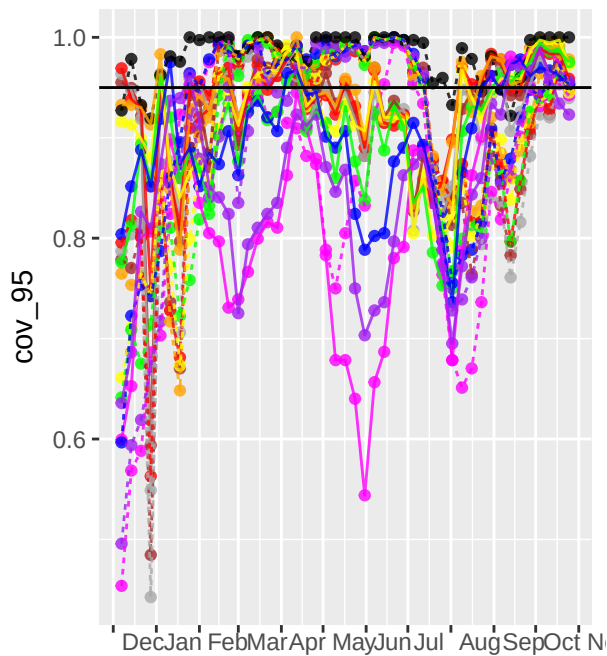
WIS (1-week)



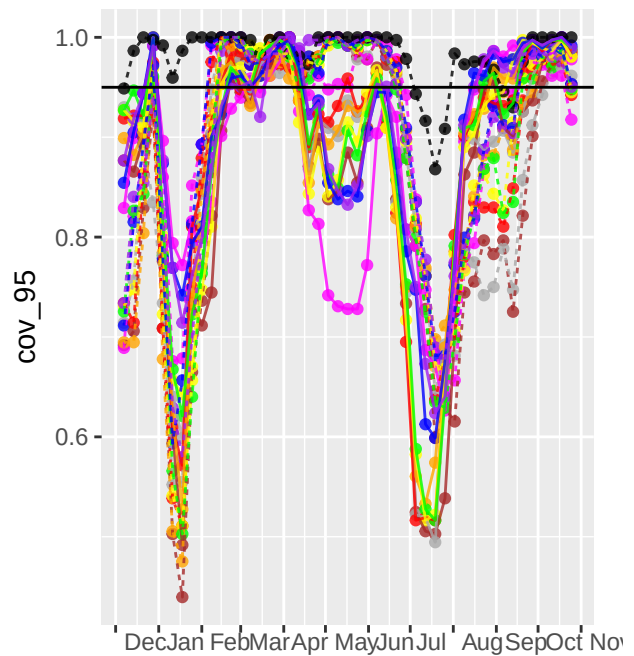
WIS (4-week)



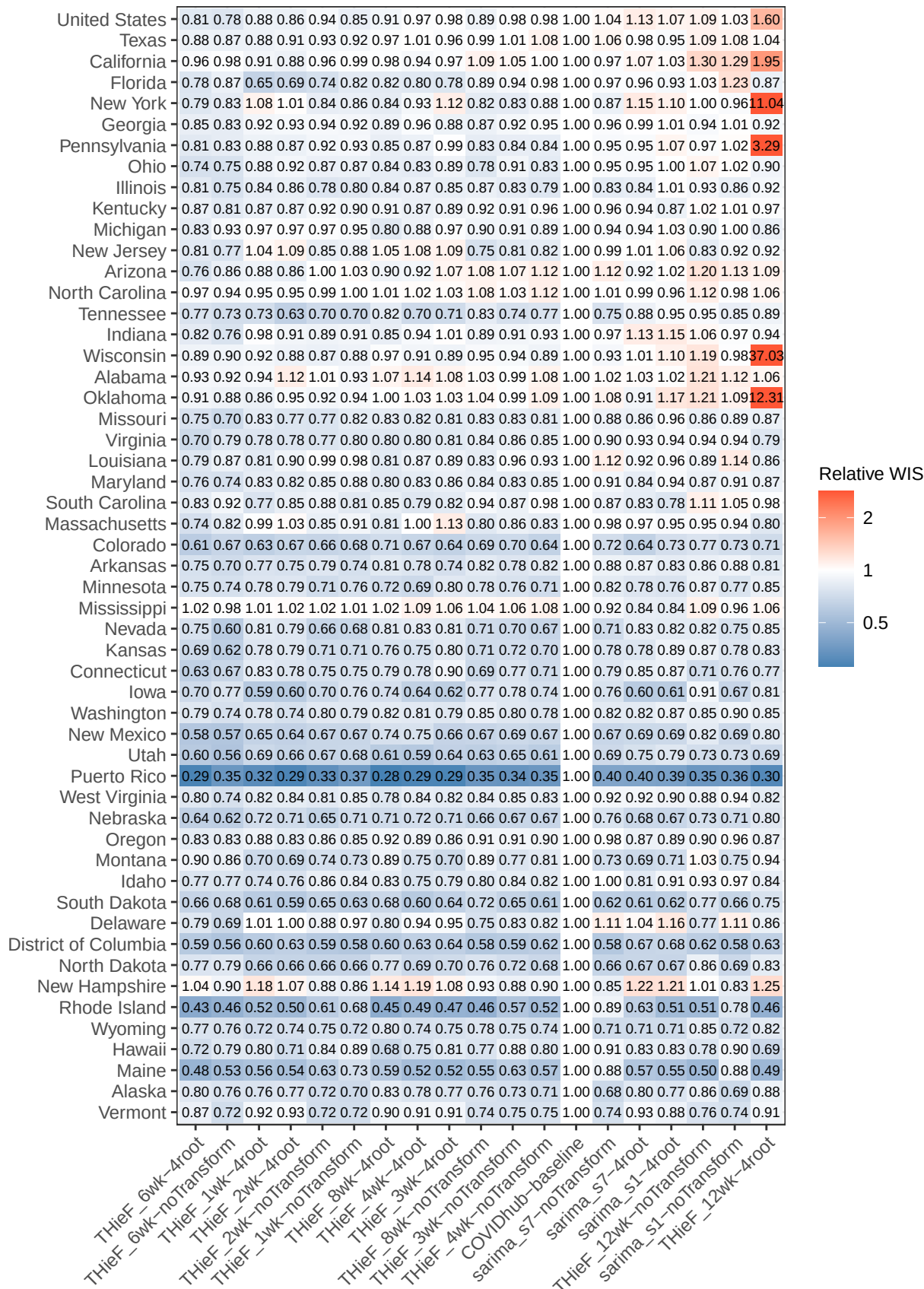
95% PI Coverage (1-week)



95% PI Coverage (4-week)



Metrics By Location



Scoring Conclusions

It seems that the best three models that should be passed onto the testing phase are the Base_arma_noTrans, the Base_arma_4root, and the Topmost8_arma_noTrans models, which seem to take turns being most accurate depending on state or national level overall and during different waves.

Original Results

Plot Forecasts

We plot forecasts for three locations (Us National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 07/05/21.

The plots don't show much difference between the 6 models (although this is partially due to the fact that the y-scales are free and are not consistent even for the same location). However, of note is the fact that the models that use a forth root transformation tend to have wider prediction intervals than the models without transformations. All of the models reasonably follow the trend of the truth lines for every location shown, although it seems like lower count locations may have slightly greater accuracy. The baseline seems to have intervals of fairly consistent widths while the models tend to vary in their interval widths, tending to show wider intervals during times of higher incidents or change points and narrow lower intervals otherwise. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

Overall metrics

Horizon week metrics

When splitting the results so that US National is separate, we find that for US National the two models that use the base set hierarchy tended to perform the best, with improvements seen over a naive baseline in all metrics except for 95%PI Coverage. These two models are followed by the Topmost8_arma_noTrans model, which showed more comparable performance to the baseline. The Base_arma_noTrans model performed the best in terms of PI Coverage, followed by the Base_arma_4root and Topmost8_arma_noTrans models (with the latter being more consistent with coverage rates for all horizons). In terms of WIS and MAE, the Base_arma_4root model performed the best, followed by the Base_arma_noTrans model. Overall, for the US National level, it seems that there is a pattern that for coverage rates, the models without transformations perform better (and these tend to have smaller prediction intervals) while for WIS and MAE the fourth root transform models perform better; these patterns hold at all horizons. Further, there is a general order to model accuracy in terms of hierarchical structure used: the Base set performs the best, followed by the Topmost8 set, and finally the Multiple3 set.

For the state level, we see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models. (We also see shrunken WIS and MAE values, although this is likely partially due to the fact that the states have smaller scale values than the National level.) Interestingly, we see fewer

improvements to MAE over the baseline but more substantial improvements in terms of coverage rates. It's a little harder to definitively say which models have the best coverage rates, as the results differ between the 50% and 95% intervals and the models are quite similar in coverage rates, especially for the 95% intervals. However, it seems that the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model.

Metrics by wave

For US National during the alpha wave, the Base_arima_noTrans model performed the best for coverage rates and second best for WIS and MAE at every horizon. The Base_arima_4root model performed best in terms of WIS and MAE but was third for coverage rate while the Topmost8_arima_noTrans model had the second best coverage rates and third best WIS and MAE. During the alpha wave, these models performed the best and tended to have more consistent values for their metrics. The winter21 wave saw noticeably worse values for all metrics at every horizon. Generally, the two models that use the base set hierarchy tended to perform the best for all metrics, although the coverage rates varied a lot by horizon, with the Topmost8_arima_4root and Multiple3_arima_noTrans achieving closer to nominal coverage rates for the 95% interval than even the Base models at horizons greater than 1 week ahead. These results are pretty consistent with improvements over the base-line, which ranged from modest to significant for the best performing models during the alpha wave, especially for WIS and 50% PI coverage. The winter21 wave saw fairly comparable or worse performance compared to the baseline for WIS and MAE. The 50% coverage was better than the baseline for all models, the greatest improvements for the best models, but the 95% coverage could be better or worse for the best models.

For states during both waves, we again see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models, compared to those of the US National level when broken into waves. During the alpha wave, the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model. During the winter21 wave, the Base_arima_4root model is again generally the best for all metrics at all horizon weeks but sometimes loses out in terms of 95% coverage at longer horizons to other models. Notably during this wave, there are much greater similarities in coverage rates and certain models have better coverage rates for the 50% interval while others have better coverage reads for the 95% one. Similar patterns to the ones described above for the US national level when compared to the baseline were observed.

By forecast date (week)

Metrics By Location

Scoring Conclusions

It seems that the best three models that should be passed onto the testing phase are the Base_arima_noTrans, the Base_arima_4root, and the Topmost8_arima_noTrans models, which seem to take turns being most accurate depending on state or national level overall and during different waves.